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# SMART SPACE

## A MAIN PROJECT REPORT

Submitted by


**TOM JOSEPH ELLICKAL**

*MLM24MCA-2059*


To

The APJ Abdul Kalam Technological University
in Partial fulfillment of the requirements for
the award of the degree

*of*

MASTER OF COMPUTER APPLICATIONS

Under the Guidance of

**Mr. Eldhose K Paul**

(Associate Professor, Department of Computer Applications)


**DEPARTMENT OF COMPUTER APPLICATIONS**

**MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR**

***(Affiliated to APJ Abdul Kalam Technological University)***


**APRIL 2026**

**MANGALAM COLLEGE OF ENGINEERING**

**Accredited by NAAC& ISO 9001:2000 Certified Institution**

**DEPARTMENT OF COMPUTER APPLICATIONS**

## VISION

To become a centre of excellence in computer applications,competent in the global ecosystem with technical knowledge,innovation with a sense of social commitment.

## MISSION

- To serve with state of the art education,foster advanced research and cultivate innovation in the field of computer applications.
- To prepare learners with knowledge skills and critical thinking to excel in the technological landscape and contribute positively to society.

## Program Educational Objectives

- PEO I :Graduates will possess a solid foundation and in-depth understanding of computer applications and will be equipped to analyze real-world problems, design and create innovative solutions, and effectively manage and maintain these solutions in their professional careers.
- PEO II: Graduates will acquire technological advancements through continued education, lifelong learning and research, thereby making meaningful contributions to the field of computing.
- PEO III: Graduates will cultivate team spirit, leadership, communication skills, ethics, and social values, enabling them to apply their understanding of the societal impacts of computer applications effectively.

## Program Specific Outcomes

- **PSO I**: Apply advanced technologies through innovations to enhance the efficiency of design development.
- **PSO II**: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

# MAPPING OF PO-PSO-SDG

## 1. MAPPING WITH PROGRAM OUTCOMES (POs):

| SL.NO | POs ADDRESSED | RELEVANCE TO PROJECT |
|---|---|---|
| 1 | PO1 | This project applies fundamental knowledge of programming, artificial intelligence, machine learning, and image processing. Concepts such as Python programming, diffusion models, and neural networks are used to build the system. |
| 2 | PO2 | The project analyzes the problem of difficulty in visualizing interior design changes. By studying user needs and limitations of traditional interior design methods, the system identifies a solution using AI-based design generation. |
| 3 | PO3 | The Career Bridge system provides an innovative solution by developing a web-based platform integrated with AI technologies. The system includes features such as AI-based resume analysis, skill gap identification, job recommendation, and mock interview practice. These functionalities help students improve their employability and assist recruiters in identifying suitable candidates. By designing and implementing these intelligent modules, the project demonstrates the development of an effective solution for modern recruitment and career guidance systems. |
| 4 | PO4 | The SmartSpace system is designed to generate interior design suggestions automatically. It develops an innovative solution using Stable Diffusion and ControlNet models to transform room images into different design styles. |
| 5 | PO8 | The project encourages learning of advanced technologies such as generative AI, diffusion models, and AI-based design tools, which supports continuous learning and adapting to new technological developments. |

**LIST OF PROGRAM OUTCOMES (POs):**

**PO1 (Foundation Knowledge):** Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving.

**PO2 (Problem Analysis):** Identify, review, formulate and analyse problems for primarily focussing on customer requirements using critical thinking frameworks.

**PO3 (Development of Solutions):** Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.

**PO4 (Modern Tool Usage):** Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

**PO5 (Individual and Teamwork):** Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.

**PO6 (Project Management and Finance):** Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.

**PO7 (Ethics):** Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware.

**PO8 (Life-long learning):** Change management skills and the ability to learn, keep up with contemporary technologies and ways of working**.**

**2. MAPPING WITH PROGRAM SPECIFIC OUTCOMES (PSOs):**

| SL.NO | PSOs ADDRESSED | RELEVANCE TO PROJECT |
|---|---|---|
| 1 | **PSO 1** | The Career Bridge system uses advanced technologies such as Python, Django, HTML, CSS, JavaScript, and AI techniques to develop an intelligent placement platform. Features like resume analysis, job recommendation, and skill gap identification improve the efficiency of recruitment and career guidance processes. |
| 2 | **PSO 2** | The project applies computing principles to analyze and solve real-world problems related to job searching and recruitment management. The system provides a practical solution by helping students find suitable jobs, build resumes, and track applications, while enabling recruiters to manage hiring processes effectively. |

**<u>LIST OF PROGRAM SPECIFIC OUTCOMES (PSOs):</u>**

**PSO 1**: Apply advanced technologies through innovations to enhance the efficiency of design development.

**PSO 2**: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

**2. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):**

| SDG NO | SDGs ADDRESSED | RELEVANCE TO PROJECT |
|---|---|---|

| | | |
|---|---|---|
| SDG 4 | Quality Education | The SmartSpace system promotes learning and skill development in advanced technologies such as Artificial Intelligence, Machine Learning, and Computer Vision**.** It also helps students and designers learn about interior design concepts through AI-generated visualizations. |
| SDG 8 | Decent Work and Economic Growth | The system can support interior designers, architects, and home planners by providing AI-based design assistance. This can improve productivity and create new opportunities in the digital design and architecture industry**.** |
| SDG 9 | Industry, Innovation, and Infrastructure | The project strongly supports this goal by applying innovative AI technologies such as Stable Diffusion and ControlNet to solve real-world problems. It promotes technological advancement in smart design systems and digital infrastructure**.** |

**SUSTAINABLE DEVLOPMENT GOALS (SDGs):**

**SDG 1 – No Poverty**-End poverty in all its forms everywhere.

**SDG 2 – Zero Hunger**-End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

**SDG 3 – Good Health and Well-Being**-Ensure healthy lives and promote well-being for all at all ages.

**SDG 4 – Quality Education**-Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

**SDG 5 – Gender Equality**-Achieve gender equality and empower all women and girls.

**SDG 6 – Clean Water and Sanitation**-Ensure availability and sustainable management of water and sanitation for all.

**SDG 7 – Affordable and Clean Energy**-Ensure access to affordable, reliable, sustainable, and modern energy for all.

**SDG 8 – Decent Work and Economic Growth**-Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

**SDG 9 – Industry, Innovation, and Infrastructure**-Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

**SDG 10 – Reduced Inequality**-Reduce inequality within and among countries.

**SDG 11 – Sustainable Cities and Communities**-Make cities and human settlements inclusive, safe, resilient, and sustainable.

**SDG 12 – Responsible Consumption and Production**-Ensure sustainable consumption and production patterns.

**SDG 13 – Climate Action**-Take urgent action to combat climate change and its impacts.

**SDG 14 – Life Below Water**-Conserve and sustainably use the oceans, seas, and marine resources.

**SDG 15 – Life on Land** -Protect, restore, and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification, halt and reverse land degradation, and halt biodiversity loss.

**SDG 16 – Peace, Justice, and Strong Institutions**- Promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable, and inclusive institutions.

**SDG 17 – Partnerships for the Goals** -Strengthen the means of implementation and revitalize the global partnership for sustainable development.

**MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR**
**DEPARTMENT OF COMPUTER APPLICATIONS**
**APRIL 2026**

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## ***DECLARATION***

*I hereby certify that the work which is being presented in the project entitled "**SMART SPACE"** submitted in the **DEPARTMENT OF COMPUTER APPLICATIONS** is an authentic record of my own work carried under the supervision of **Mr. Eldhose K Paul**, **Associate professor**. This study has not been submitted to any other institution or university for the award of any other degree. This report has been checked for plagiarism by the college and the similarity index is within permissible limits set by the college.*

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*TOM JOSEPH ELLICKAL*

*Date:*

*Place:*

**MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR**
**DEPARTMENT OF COMPUTER APPLICATIONS**
**APRIL 2026**

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## ***CERTIFICATE***

Certified that the report entitled **'SMART SPACE'** submitted by **TOM JOSEPH ELLICKAL (MLM24MCA-2059)** to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the **DEGREE OF MASTER OF COMPUTER APPLICATIONS** by **APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY** is a bonafide record of the MAIN PROJECT work carried out by him under our guidance and supervision during the academic year 2025-26.

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# ABSTRACT

The Smart space interior design plays an important role in improving the comfort, functionality, and visual appeal of living spaces. However, traditional interior design methods often require professional expertise, significant time, and high cost. With the advancement of artificial intelligence and generative models, it has become possible to automate and simplify the design process. This project presents SmartSpace, an AI-driven interior design system that generates creative and realistic room design suggestions based on user input.

The proposed system allows users to upload an image of their room and provide design preferences such as interior style, color theme, and layout ideas. The system uses advanced deep learning models, specifically Stable Diffusion and ControlNet, to analyze the room structure and generate multiple interior design variations while preserving the original layout of the room. ControlNet ensures that the structural elements such as walls, windows, and room boundaries remain consistent, while Stable Diffusion generates visually appealing interior designs based on the given prompts.The system is implemented through a web-based interface where users can easily upload images, select design preferences, and view the generated design results. The generated designs are stored in a database so that users can access and download them later. By combining artificial intelligence with user-friendly design tools, the system helps users visualize different interior styles before making real-world modifications.The results demonstrate that the system can effectively generate multiple creative design suggestions, helping users explore various interior design possibilities quickly and efficiently. The SmartSpace system reduces the effort required for planning interior spaces and provides a convenient tool for homeowners, designers, and architects. In the future, the system can be enhanced by integrating features such as real-time design editing, augmented reality visualization, and personalized furniture recommendations.



***Keywords: AI-Based Interior Design, Stable Diffusion, ControlNet, Image-to-Image Generation, Smart Room Design, Deep Learning, Generative AI, Interior Visualization, Automated Design System, Computer Vision.***


| | |
|---|---|
| **Mapping with Sustainable Development Goals (Mention the Goals)** | **SDG 4 – Quality Education**<br>**SDG 8 – Decent Work and Economic Growth**<br>**SDG 9 – Industry, Innovation, and Infrastructure** |

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# LIST OF ABBREVIATIONS

| ABBREVIATION | | FULL FORM |
|---|---|---|
| AR | - | Augmented Reality |
| AI | - | Artificial Intelligence |
| API | - | Application Programing interface |
| CNN | - | Convolutional neural networks |
| UX | - | User experience |
| CN | - | ControlNet |

# CHAPTER 1
# INTRODUCTION

## 1.1 Background

Artificial Intelligence (AI) is transforming many industries by automating complex tasks and improving decision-making processes. One of the emerging applications of AI is in the field of interior design, where technology can assist in creating visually appealing and functional living spaces. Traditional interior design often requires significant time, expertise, and resources to generate suitable layouts and design ideas. With the advancement of machine learning and generative models, it has become possible to automate and enhance this process.

This project presents an **AI-driven interior design system** that uses the **Stable Diffusion model** to generate personalized room layouts and design recommendations. The system analyzes user preferences and room images to produce creative and practical interior design suggestions. By leveraging artificial intelligence techniques, the system can automatically generate different room styles such as modern, minimalist, and rustic, allowing users to visualize multiple design possibilities.

The proposed system aims to simplify the interior design process by providing customized design outputs based on room dimensions, layout structure, and user inputs. This helps homeowners and designers explore various design options quickly and efficiently without requiring extensive manual planning. The system also enables users to visualize potential transformations in their spaces before implementing them in real life.

Overall, this project demonstrates how AI can be effectively integrated into interior design to improve creativity, efficiency, and accessibility. By combining advanced machine learning models with user-friendly design generation, the system offers a modern solution for creating personalized interior environments.

## 1.2 Motivation

Designing an interior space is an important part of creating a comfortable and attractive living environment. However, many people find it difficult to plan room layouts, choose suitable furniture, or decide on the right design style. Traditional interior design usually requires professional designers, which can be expensive and time-consuming. Because of this, many homeowners struggle to visualize how their rooms will look after making design changes.

With the growth of modern technologies, especially **Artificial Intelligence (AI)**, it has become possible to simplify many creative tasks. AI-based image generation models can analyze visual information and produce new design ideas automatically. These technologies provide an opportunity to support users by generating design suggestions quickly and efficiently without requiring advanced design knowledge.

The motivation behind this project is to develop a system that uses AI to help users explore different interior design possibilities. By using an AI model capable of generating images and layouts, users can upload room photos and receive design recommendations based on different styles and preferences. This allows users to experiment with multiple design options before making real changes to their space.

Another motivation for this project is to make interior design more accessible for everyone. Instead of relying only on professional designers, people can use an intelligent system to visualize their ideas and improve their living spaces. The system can also help designers by providing inspiration and reducing the time needed for early design planning. These technologies provide an opportunity to support users by generating design suggestions quickly and efficiently without requiring advanced design knowledge.Overall, this project is motivated by the goal of combining **creativity, technology, and automation** to make the interior design process easier, faster, and more interactive for users. The system can also help designers by providing inspiration and reducing the time needed for early design planning. These technologies provide an opportunity. The system can also help designers by providing inspiration and reducing the time needed for early design planning. These technologies provide an opportunity.

### 1.3 Scope and Objective

The main objective of this project is to develop an intelligent interior design system that uses Artificial Intelligence to generate personalized room layouts and design suggestions. The system aims to simplify the interior design process by allowing users to visualize different design ideas automatically. Instead of relying only on manual planning or professional designers, users can use this system to explore various design possibilities in a faster and more convenient way.

One of the key objectives of the project is to integrate an AI-based image generation model that can analyze room images and produce creative interior design outputs. By using advanced machine learning techniques, the system can generate different types of room designs based on user preferences and input data. This allows users to experiment with multiple design styles and select the most suitable option for their space.

Another important objective is to provide a user-friendly platform where users can easily interact with the system. The interface should allow users to upload room images, select preferred design styles, and receive design recommendations. The goal is to make the system simple enough for both professionals and non-experts to use without requiring specialized design knowledge.

The project also aims to support multiple interior design styles such as modern, minimalist, and rustic. By offering a variety of design options, the system can address different aesthetic preferences and provide users with more flexibility when planning their interior spaces.

In addition, the system aims to improve efficiency in the design process. Traditional interior design often involves several stages of planning, visualization, and modification. By automating these steps with AI technology, the system can significantly reduce the time required to generate design ideas and layouts.

Another objective is to help users visualize potential changes in their living spaces before implementing them in real life. Visualization plays an important role in interior design because it

helps users understand how furniture placement, color combinations, and decorations will look together. The system provides generated images that represent possible room transformations.

Finally, the project aims to demonstrate the practical application of Artificial Intelligence in creative industries. By combining machine learning techniques with interior design concepts, the system highlights how AI can support innovation, improve productivity, and assist users in making better design decisions.

The scope of this project focuses on the development of an AI-driven system that assists users in generating interior design ideas and room layouts. The system is designed to analyze room images and user preferences to produce visually appealing and practical design suggestions. This project mainly focuses on applying Artificial Intelligence techniques to automate and simplify the early stages of interior design planning.

The system allows users to upload images of their rooms and receive AI-generated design recommendations. Based on the provided input, the system can generate different design styles and layout variations. This helps users explore multiple interior design possibilities without the need for complex design tools or professional software.

The project primarily supports common residential room types such as bedrooms, living rooms, dining rooms, and kitchens. For each type of room, the system can generate design ideas that include furniture arrangements, color themes, and decoration styles. This enables users to visualize how different elements can be combined to create a balanced and attractive interior space.

Another part of the project's scope is the integration of a machine learning model capable of producing realistic design outputs. The system focuses on generating visual representations of interior spaces that can help users better understand the potential appearance of their rooms after redesigning them.

The system is also designed to support multiple interior design styles. Users can select different styles based on their personal preferences, allowing them to explore a variety of design themes. This feature improves flexibility and makes the system suitable for different user needs and aesthetic tastes.

In addition, the project aims to provide a simple and interactive interface where users can easily access the system's features. The goal is to ensure that the platform can be used by homeowners, students, and designers who want quick design inspiration without complex technical knowledge.

### 1.4 Problem Statement

Interior design is an important aspect of creating comfortable, functional, and visually appealing living spaces. However, designing a room layout and selecting appropriate furniture, colors, and decorative elements can be a challenging task for many people. Most individuals find it difficult to imagine how different design elements will look together before actually implementing them. This lack of visualization often leads to poor design choices, wasted resources, and dissatisfaction with the final outcome.

Traditional interior design typically depends on professional designers who possess the expertise to plan and visualize spaces effectively. While professional services provide quality results, they can be expensive and time-consuming. Many homeowners and individuals may not have access to professional interior designers due to financial limitations or availability. As a result, people often rely on trial-and-error methods or online inspiration, which may not always provide personalized solutions for their specific room structures and preferences.

Another challenge is that existing interior design tools and software often require technical knowledge and design experience. Many platforms are complex and difficult for non-expert users to operate. These tools may also lack the ability to automatically generate design ideas based on user input and room characteristics. Therefore, users may still struggle to create practical and aesthetically pleasing layouts without professional guidance.

With the advancement of Artificial Intelligence and machine learning technologies, there is an opportunity to develop intelligent systems that can assist users in generating creative design ideas automatically. AI-based models have the ability to analyze images and generate new visual content, which can be applied to interior design to create personalized room layouts and design suggestions.

However, there is still a need for a system that can effectively combine AI image generation with user-friendly interaction to help users visualize different interior design possibilities. Such a system should be capable of analyzing room photos, understanding user preferences, and generating multiple design styles that can assist users in making better design decisions.

Therefore, the main problem addressed in this project is the lack of accessible, intelligent, and automated tools that help users easily visualize and generate personalized interior design layouts based on their room structure and preferences**.** This project aims to solve this problem by developing an AI-driven interior design system that uses advanced image generation techniques to provide customized design recommendations and improve the overall design planning process.

# CHAPTER 2
# REQUIREMENT ANALYSIS

## 2.1 Existing System and Problem Statement

Interior design has traditionally been carried out by professional designers who use their experience and creativity to plan room layouts, choose furniture, and select color combinations. In the existing system, people usually depend on interior designers, design magazines, or online images for inspiration when planning the design of their homes. Designers typically create sketches, digital models, or 3D visualizations to present design ideas to their clients.

Although this traditional approach can produce high-quality results, it also has several limitations. Hiring professional designers can be expensive, and the design process may take a considerable amount of time. Many homeowners cannot afford professional design services or may not have easy access to experienced designers.

Another existing approach is the use of interior design software and online tools. Some software applications allow users to create room layouts and experiment with furniture placement. However, these tools often require technical knowledge and design skills to operate effectively. Beginners may find these applications complicated and difficult to use. Additionally, many tools rely heavily on manual input, meaning the user must design the layout step by step without much intelligent assistance.

Most existing systems also lack the ability to automatically generate personalized design ideas based on the user's room images and preferences. As a result, users still need to spend a significant amount of time searching for design inspiration and manually experimenting with different arrangements.

Despite the availability of interior design tools and professional services, many people still face difficulties in planning and visualizing their interior spaces. One of the main challenges is the

inability to clearly imagine how different design elements such as furniture placement, color themes, and decorations will appear together in a real room.

Another major problem is that professional interior design services are often costly and time-consuming. This makes it difficult for many individuals to access personalized design guidance. At the same time, existing software tools require users to have some level of design knowledge, which may not be suitable for beginners.

In addition, many existing design platforms do not use advanced Artificial Intelligence techniques to automatically generate creative design suggestions. As a result, users must rely heavily on manual work and external inspiration sources.

Therefore, there is a need for an intelligent system that can assist users in generating interior design ideas quickly and efficiently. The system should be able to analyze room images, understand user preferences, and generate visually appealing design suggestions automatically.

This project addresses the problem by developing an **AI-based interior design system** that uses advanced image generation techniques to create personalized room layouts and design recommendations. The proposed system aims to make interior design more accessible, efficient, and user-friendly for a wider range of users.

### 2.2 Solution Overview

The proposed solution focuses on developing an intelligent interior design system that uses Artificial Intelligence to assist users in generating creative and personalized room designs. The system is designed to simplify the interior design process by automatically producing design ideas based on user input and room images. By combining AI technology with a user-friendly interface, the system helps users visualize different design possibilities without requiring professional design experience.

The core component of the system is an AI-based image generation model that can analyze room images and produce redesigned versions of the same space with different interior styles. The user

can upload an image of their room through the application interface. Once the image is uploaded, the system processes the image and prepares it for analysis. After processing, the AI model generates interior design suggestions based on selected styles or user preferences.

The system allows users to choose from various interior design themes such as modern, minimalist, traditional, or contemporary. Based on the selected style, the AI model generates images that represent how the room could look after redesigning. These generated images provide a visual representation of furniture arrangements, color combinations, and decorative elements that match the chosen theme.

Another important part of the solution is the system interface. The platform is designed to be simple and easy to use so that users can interact with it without technical difficulty. The interface provides options for uploading images, selecting design styles, and viewing the generated design results. This allows users to quickly explore multiple design ideas and compare different styles.

The solution also aims to improve the efficiency of the design process. Instead of manually creating multiple design layouts, the AI model can generate several design variations within a short period of time. This helps users save time and effort while exploring interior design options.

In addition, the system provides a flexible and scalable platform that can be expanded in the future. Additional features such as more design styles, furniture recommendations, and advanced customization options can be integrated to further enhance the user experience.

Overall, the proposed solution uses Artificial Intelligence to create a smart interior design assistance system. By automating the generation of design ideas and providing visual outputs, the system helps users make better design decisions and simplifies the overall interior planning process

### 2.3 Feasibility Study

A feasibility study is an important stage in system development that helps determine whether a proposed project is practical and achievable. It evaluates different aspects of the system to ensure that the project can be successfully developed and implemented. The feasibility study examines

factors such as technical capability, economic cost, operational usability, and time requirements. For the proposed AI-based interior design system, the feasibility study helps

determine whether the system can be developed effectively using available resources and technologies.

### 2.3.1 Technical Feasibility

Technical feasibility evaluates whether the required technology, tools, and technical knowledge are available to develop the proposed system. The AI-driven interior design system can be implemented using existing machine learning frameworks and programming languages such as Python. Modern AI models, including image generation models, provide the capability to analyze images and generate new visual outputs.

The development environment can include software tools such as Visual Studio Code and frameworks such as Flask or Django for building the web application. These technologies are widely used and well supported, making them suitable for developing the system. Additionally, the required libraries and AI models are available through open-source platforms, which makes the technical implementation feasible.

Since the system mainly focuses on generating interior design images based on user input, it does not require extremely complex hardware infrastructure. A computer or laptop with sufficient processing capability and internet connectivity is enough to develop and run the system. Therefore, the project is technically feasible

### 2.3.2 Economic Feasibility

Economic feasibility examines whether the project can be developed within a reasonable budget. The proposed system does not require expensive proprietary software or specialized equipment. Most of the development tools and frameworks used in the project are open-source and freely available. Programming languages such as Python and libraries used for AI development are available at no cost. Similarly, development tools like Visual Studio Code are also free to use.

Because the project mainly relies on software resources rather than costly hardware or commercial tools, the overall development cost remains low.As a result, the project can be implemented without significant financial investment, making it economically feasible for academic development and experimentation.

### 2.3.3 Operational Feasibility

Operational feasibility evaluates how well the proposed system will function in a real-world environment and whether users will be able to use it effectively. The AI-based interior design system is designed with a simple and user-friendly interface so that users can interact with it easily.Users only need to upload room images and select their preferred design styles. The system then generates design suggestions automatically. This simple workflow makes the system suitable for both beginners and users with limited technical knowledge.The system can be used by homeowners, students, or designers who want to explore different interior design ideas quickly. Because of its ease of use and automated functionality, the system is operationally feasible.

# CHAPTER 3

# LITERATURE REVIEW

## 3.1 Stable Diffusion for Text-to-Image Generation (2022) By Robin Rombach et al. (CVPR)

In 2022, Robin Rombach and his research team introduced **Stable Diffusion**, a powerful deep learning model designed for generating high-quality images from textual descriptions. The research was presented at the Conference on Computer Vision and Pattern Recognition. The primary goal of this work was to develop an efficient image generation model capable of producing realistic images while reducing computational requirements.

The proposed model uses a **latent diffusion technique**, where images are generated in a compressed latent space rather than directly in high-resolution pixel space. This approach significantly reduces memory usage and improves the speed of image generation. The model was trained on large image datasets, allowing it to understand visual patterns such as shapes, colors, textures, and spatial arrangements.

Stable Diffusion enables users to generate images based on text prompts, making it highly useful for creative applications such as art generation, graphic design, and interior design visualization. By providing descriptive prompts, users can create different visual outputs that match specific themes or styles.

The study demonstrated that diffusion-based models can generate detailed and realistic images compared to many previous generative models. This technology has opened new opportunities for developing intelligent systems that assist users in generating design ideas automatically.

The research provides a strong foundation for AI-based image generation systems. In the context of the proposed project, Stable Diffusion can be used to generate interior design layouts and room visualization based on user preferences and room images.

### 3.2 CLIP: Connecting Text and Images (2021) By Alec Radford et al. (OpenAI)

In 2021, Alec Radford and his research team introduced **CLIP (Contrastive Language–Image Pretraining)** at OpenAI. The study focused on connecting natural language descriptions with visual images using deep learning models. The main objective of this research was to train an AI system capable of understanding images based on textual descriptions.

The CLIP model was trained using a large dataset containing images paired with text captions. Through this training process, the system learned the relationship between language and visual features. As a result, the model can identify objects in images and match them with relevant text descriptions.

This technology has been widely used in modern AI applications such as image classification, visual search, and image generation. When combined with image generation models, CLIP helps improve the accuracy of generated images by ensuring they match the user's textual input.

The research contributed significantly to the development of AI-powered creative systems. In interior design applications, CLIP can help generate images that match user descriptions of room styles, furniture arrangements, and design themes.

### 3.3 Generative Adversarial Networks for Image Generation (2014) By Ian Goodfellow et al. (NIPS)

In 2014, Ian Goodfellow and his colleagues introduced **Generative Adversarial Networks (GANs)** at the Neural Information Processing Systems Conference. The research proposed a new deep learning framework capable of generating realistic images through a competitive learning process between two neural networks.GANs consist of two main components: a generator and a discriminator. The generator creates images, while the discriminator evaluates whether the generated images are real or fake. Through continuous training, the generator improves its ability to produce realistic images that closely resemble real-world data.

This technology has been widely used in image generation tasks, including art generation, photo enhancement, and virtual design visualization. GAN-based models have also been applied in architectural and interior design fields to generate creative design concepts.The research demonstrated how AI can automatically create visual content, which inspired many modern AI-based design systems.

### 3.4 High-Resolution Image Synthesis Using StyleGAN (2019) By Tero Karras et al. (NVIDIA Research)

In 2019, Tero Karras and his team from NVIDIA Research introduced **StyleGAN**, an advanced generative model designed for producing high-resolution images. The study focused on improving the quality and realism of AI-generated images. StyleGAN introduced a new architecture that allows better control over different visual features such as color, texture, and structure. This approach allows users to modify specific aspects of generated images while maintaining overall image quality.The model demonstrated impressive results in generating realistic human faces and other visual objects. The research also showed that generative models could be used to produce detailed visual designs for creative applications. In the context of interior design systems, StyleGAN-based techniques can assist in generating detailed room visualizations and design variations.

### 3.5 Diffusion Models Beat GANs on Image Synthesis (2021) By Prafulla Dhariwal and Alex Nichol (OpenAI)

In 2021, Prafulla Dhariwal and Alex Nichol proposed diffusion-based models for image synthesis at OpenAI. Their research showed that diffusion models could generate higher-quality images compared to many GAN-based models.Diffusion models work by gradually adding noise to images during training and then learning how to reverse this process to generate new images. This method helps produce detailed and realistic outputs.The study demonstrated that diffusion models provide better stability and improved image quality compared to earlier generative models. As a result, diffusion-based approaches became widely used in modern AI image generation systems.This research laid the foundation for many recent image generation technologies used in design and visualization systems.

### 3.6 Computer-Aided Interior Design Systems (2018) By Xiangyu Zhang et al. (IEEE)

In 2018, Xiangyu Zhang and his research team studied computer-aided interior design systems. Their research focused on using digital tools to assist designers in planning room layouts and furniture arrangements. The proposed system used 3D modeling techniques to visualize interior spaces and allow users to experiment with different design options. Users could adjust furniture placement, color schemes, and lighting arrangements within the digital environment. Although the system provided helpful visualization features, it required manual interaction and design knowledge from users. The research highlighted the need for more intelligent systems that could automatically generate design ideas.

### 3.7 AI-Based Interior Design Recommendation System (2023) By Li Wang et al. (IEEE Access)

In 2023, Li Wang and colleagues developed an AI-based recommendation system for interior design. The main objective of this research was to provide automated design suggestions based on user preferences and room characteristics. The system analyzed user input such as preferred color schemes, furniture styles, and room types. Using machine learning algorithms, the system recommended suitable design elements that matched the user's preferences. The research showed that AI can significantly assist users in selecting appropriate interior design features without requiring professional design skills. The system helped users visualize different design possibilities before implementing changes in real spaces.

### 3.8 Deep Learning for Room Layout Generation (2020) By Kai Chen et al. (ACM)

In 2020, Kai Chen and his team proposed a deep learning approach for generating room layouts automatically. The study focused on using neural networks to analyze spatial relationships between furniture and interior objects.The system was trained using datasets containing different room layouts and furniture arrangements. Through training, the model learned patterns related to how furniture is typically placed in real interior spaces.

The generated layouts helped users visualize how furniture could be arranged in a room while maintaining functional and aesthetic balance. This research demonstrated the potential of deep learning techniques in automating interior design planning.

### 3.9 Text-to-Image Generation Using DALL·E (2021) By Aditya Ramesh et al. (OpenAI)

In 2021, Aditya Ramesh and his team at OpenAI introduced **DALL·E**, an AI system capable of generating images from textual descriptions.The model uses transformer-based neural networks to interpret text prompts and convert them into visual representations. Users can describe a scene or object using natural language, and the system generates an image that matches the description.This research demonstrated the powerful capabilities of AI in creative visual generation tasks. The technology has been widely used in design, art generation, advertising, and concept visualization.In interior design applications, such models can generate room designs based on textual descriptions provided by users.

### 3.10 AI-Assisted Interior Visualization Systems (2022) By Maria Gonzalez et al. (IEEE)

In 2022, Maria Gonzalez and her research team studied AI-assisted interior visualization systems. Their research focused on integrating computer vision techniques with AI image generation models. The system analyzed images of existing rooms and generated alternative interior design concepts. These generated designs allowed users to explore different decoration styles and furniture arrangements. The study highlighted the benefits of using artificial intelligence to support creative design processes. AI-assisted visualization systems can help users better understand how their living spaces may look after redesigning.This research demonstrated that AI can improve creativity, efficiency, and accessibility in interior design planning.

# CHAPTER 4

# DIAGRAMS

System diagrams play an essential role in understanding the structure, workflow, and interaction between different components of a software system. These diagrams provide a visual representation of how the Fake News Detection and Visualization Platform operates internally. By using diagrams such as Data Flow Diagrams (DFD), Use Case Diagrams, Class Diagrams, State Diagrams, Activity Diagrams, and Sequence Diagrams, developers and users can easily understand the flow of information and the functionality of each system component.The diagrams presented in this chapter describe the architecture and behavior of the proposed fake news detection system. They illustrate how the system processes news input provided by users, how machine learning models analyze the data, and how results are generated and visualized.These diagrams also help in identifying system modules, data storage mechanisms, and interactions between system components. Each diagram provides a different perspective of the system design, ranging from high-level system interactions to detailed internal workflows.

## 4.1 Data flow Diagram (DFD)

A Data Flow Diagram represents how data moves through the system. It shows the input, processing steps, and output of the system. In this project, the DFD explains how the room image and design preferences provided by the user are processed by the system. The processed data is then sent to the AI model, which generates interior design suggestions. DFDs help in clearly understanding the flow of information inside the system.

### 4.1.1 DFD Level 0

**Data Flow Diagram (DFD) Level 0** is also called a **Context Diagram**. It represents the entire system as a single process and shows how external entities interact with the system. It gives a high-level view of how data enters the system, how it is processed, and how the output is delivered.

In this project, the **AI-Based Interior Design System** acts as the main process. The user is the primary external entity that interacts with the system. The user provides inputs such as room images or design preferences through the application interface. The system processes this information using AI models and generates interior design suggestions as output.

The main purpose of the Level 0 DFD is to provide a simple overview of the system without going into detailed internal processes. It helps in understanding the relationship between the user and the system and how data flows between them.

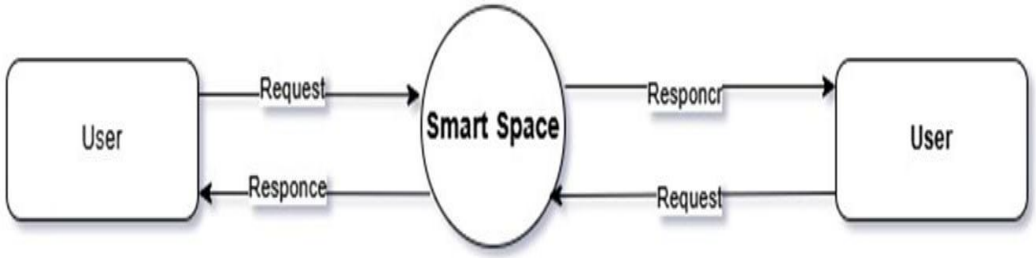


Figure 4.1.1 DFD Level 0

The diagram represents the **Level 0 Data Flow Diagram (Context Diagram)** of the **Smart Space / AI Interior Design System**. It provides a high-level overview of how users interact with the system and how requests and responses flow between them.At the center of the diagram is the **Smart Space system**, which acts as the main processing unit. This system receives input from

users, processes the information, and returns the appropriate results.On both sides of the diagram there are **User entities**. These users represent individuals who interact with the system to obtain interior design suggestions or space-related outputs.

The interaction begins when a **user sends a request** to the Smart Space system. This request may include actions such as uploading a room image, selecting a design style, or requesting design recommendations. The system receives this request and processes it using its internal modules and AI-based design generation mechanisms. After processing the request, the **Smart Space system generates a response**. The response may include generated interior design images, layout suggestions, or design recommendations based on the user's input.The arrows in the diagram represent the **flow of data between the user and the system**:

- **Request:** Data or instructions sent from the user to the Smart Space system.
- **Response:** Output generated by the system and returned to the user.

This Level 0 DFD focuses only on the **overall interaction between the user and the system**, without showing internal processes. It helps in understanding the general functionality of the system and how information flows between the user and the Smart Space application.

### 4.1.2 DFD LEVEL 1

A **DFD Level 1** diagram provides a more detailed view of the system compared to Level 0. While Level 0 shows the system as a single process, Level 1 breaks the main system into multiple internal processes and shows how data flows between them. It helps in understanding how different modules inside the system work together to complete the task.In the **AI-Based Interior Design System (Smart Space)**, the Level 1 DFD explains the internal processes that occur after the user sends a request to the system. The user first interacts with the system by uploading a room image or providing a design request. This input is received by the system and passed to different processing modules.The first process is **Image/Input Collection**, where the system receives the room image or design preference from the user. After that, the data moves to the **Image Processing Module**, where the uploaded image is analyzed and prepared for the AI model. This step may

include resizing the image or extracting important visual features.Next, the processed data is sent to the **AI Model (Stable Diffusion)**. This module is responsible for generating new interior design layouts based on the input image and selected design style. The AI model analyzes the input and

produces creative interior design outputs.Once the design is generated, the results are passed to the **Design Output Module**, where the generated interior design images are prepared for display. Finally, the system sends the generated design results back to the **user** through the interface.

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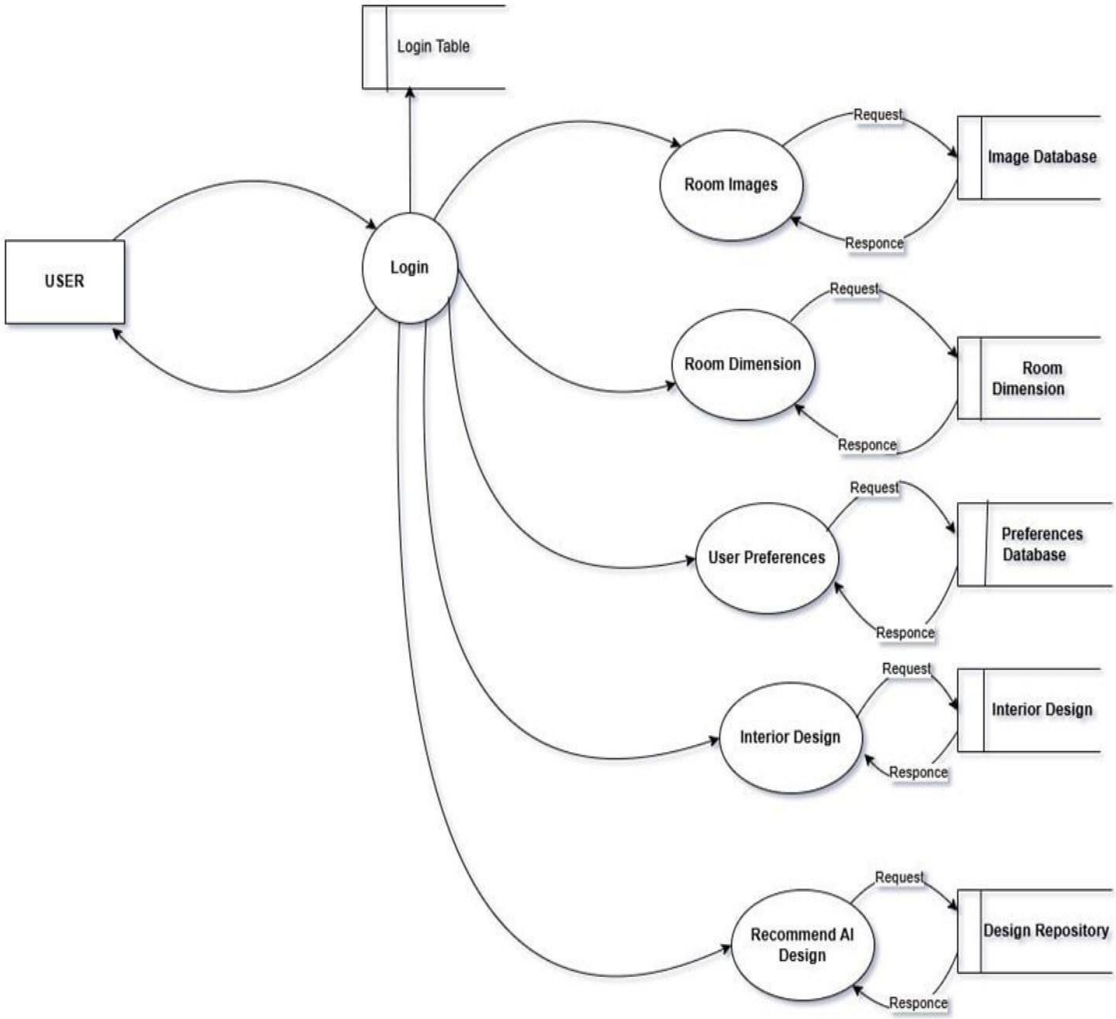



Figure 4.1.2 DFD LEVEL 1

**1. User**

The **User** is the external entity who interacts with the system. The user starts the process by accessing the application and entering login details. After logging in, the user can upload room

images, provide room dimensions, and specify design preferences to receive AI-based interior design suggestions.

**2. Room Images Process**

After logging in, the user can upload images of the room that needs interior design suggestions. The **Room Images** process collects and manages these images. The system sends a request to the **Image Database** to store or retrieve the room images. Once processed, the database sends a response back to the system.

**3. Room Dimension Process**

The **Room Dimension** process handles information related to the size and layout of the room. The user provides details such as room length, width, or other measurements. The system sends a request to the **Room Dimension database** to store or retrieve this information. The database then returns the required response to the system.

**4. User Preferences Process**

In this process, the system collects user preferences related to interior design styles. The user may select options such as modern, minimalist, traditional, or rustic styles. These preferences are stored in the **Preferences Database**. The system sends requests to the database and receives responses containing stored user preference information.

**5. Interior Design Process**

The **Interior Design** process generates design ideas based on the room images, dimensions, and user preferences. The system processes all the collected information and sends requests to the **Interior Design database** for design resources or templates. The database responds with relevant data that helps generate the final design concept

**6. Recommend AI Design Process**

This process represents the **AI-based recommendation module** of the system. The system analyzes all the input data and uses artificial intelligence to generate design suggestions. It interacts with the **Design Repository**, where design templates or generated designs are stored. The repository sends responses containing recommended interior design layouts.

### 4.2 Use Case Diagram

A Use Case Diagram describes the interaction between users and the system. It identifies the different actions that users can perform within the system. In this project, the user can upload room images, select interior design styles, and view generated design results. This diagram helps define the system's functionality from the user's perspective.

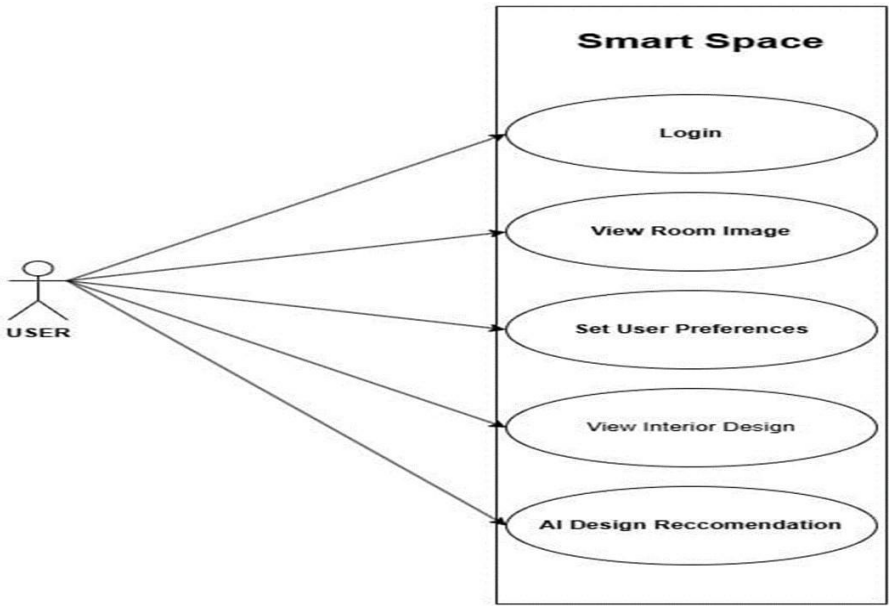



Figure 4.2 Use Case Diagram

The diagram represents the Use Case Diagram of the Smart Space AI Interior Design System. A use case diagram describes how a user interacts with a system and what functions the system provides. In this diagram, the User is the main actor who interacts with different features of the Smart Space system.

**1. User (Actor)**

The **User** is the external entity who uses the Smart Space application. The user interacts with the system to upload room images, set design preferences, and obtain AI-generated interior design suggestions. All the system functionalities are accessed by the user through the interface.

**2. Login**

The **Login** use case allows the user to access the Smart Space system securely. The user enters login credentials such as username and password. The system verifies these credentials and grants access to the application. This step ensures that only authorized users can use the system features.

**3. View Room Image**

The **View Room Image** function allows the user to upload or view images of their room. These images are used by the system to understand the existing room layout and structure. The user can check the uploaded images before generating design suggestions.

**4. Set User Preferences**

In this use case, the user specifies their design preferences. This may include selecting interior design styles such as **modern, minimalist, traditional, or rustic**, along with color schemes or furniture types. These preferences help the system generate more personalized interior design suggestions.

**5. View Interior Design**

The **View Interior Design** feature allows the user to see the generated interior design layouts. After processing the user inputs, the system provides visual outputs or design suggestions. The user can review these results to understand how the room can be redesigned. user to see the generated interior design layouts. After processing the user inputs, the system provides visual outputs or design suggestions

**6. AI Design Recommendation**

This use case represents the **core feature of the system**. The Smart Space system uses artificial intelligence to analyze the uploaded room images, dimensions, and user preferences. Based on this information, the system generates suitable interior design recommendations. These AI-generated designs help users visualize potential improvements for their room.

## 4.3 Class Diagram

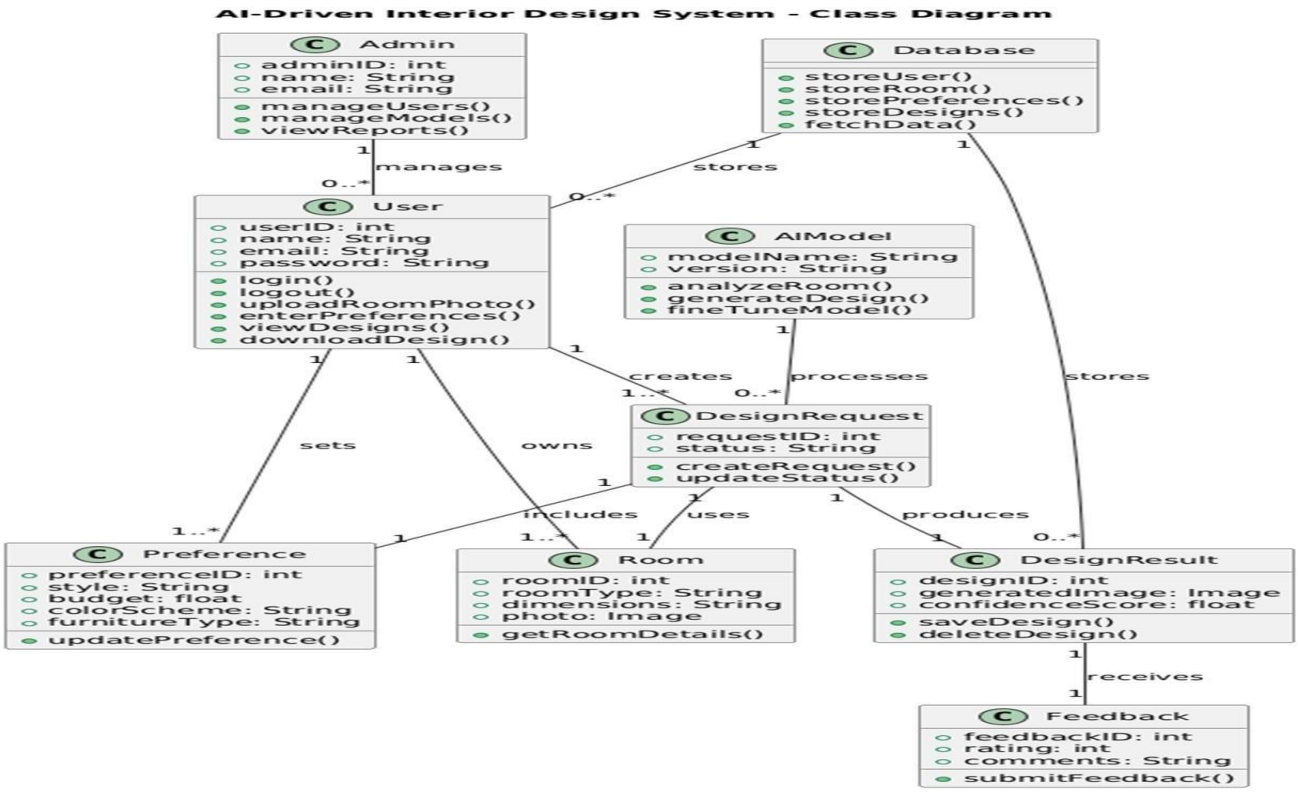



Figure 4.3 class Diagram

The **Class Diagram** represents the structure of the AI-Driven Interior Design System. It shows the main classes in the system, their attributes, functions (methods), and the relationships between them. Each class represents a component of the system that helps in generating AI-based interior design recommendations.

### 1. Admin Class

The **Admin** class represents the administrator who manages the system.

**Attributes**

- adminID : int – Unique ID of the administrator
- name : String – Administrator name
- email : String – Administrator email

**Methods**

- manageUsers() – Manages user accounts in the system
- manageModels() – Manages the AI models used in the system
- viewReports() – Views system reports and usage data

**Relationship**

- One admin can **manage multiple users** in the system.

### 2. User Class

The **User** class represents the person who uses the application to generate interior design suggestions.

**Attributes**

- userID : int – Unique user identification number
- name : String – Name of the user
- email : String – Email address of the user
- password : String – Password used for authentication

**Methods**

- login() – Allows the user to log into the system
- logout() – Logs the user out of the system
- uploadRoomPhoto() – Uploads a room image
- enterPreferences() – Sets design preferences
- viewDesigns() – Displays generated designs
- downloadDesign() – Allows the user to download the design

**Relationship**

- A user can **create design requests**.
- A user **sets preferences** for design generation.
- A user **owns room information**.

**3. Preference Class**

The **Preference** class stores the design preferences provided by the user.

**Attributes**

- preferenceID : int – Unique preference ID
- style : String – Interior design style
- budget : float – Budget limit for design
- colorScheme : String – Preferred color scheme

- furnitureType : String – Preferred furniture style

**Method**

- updatePreference() – Updates the design preference information.

**Relationship**

- Each user can **set one or more preferences**.

**4. Room Class**

The **Room** class represents the room information used for generating designs.

**Attributes**

- roomID : int – Unique room identification
- roomType : String – Type of room (bedroom, kitchen, etc.)
- dimensions : String – Room size details
- photo : Image – Image of the room

**Method**

- getRoomDetails() – Retrieves information about the room.

**Relationship**

- The room data is **used by the design request process**.

**5. DesignRequest Class**

The **DesignRequest** class represents a request created by the user to generate an interior design.

**Attributes**

- requestID : int – Unique ID for the design request
- status : String – Current status of the request

**Methods**

- createRequest() – Creates a new design request
- updateStatus() – Updates the request status

**Relationship**

- A user **creates design requests**.
- The request **uses room and preference data**.
- The request is processed by the **AI Model**.

**6. AIModel Class**

The **AIModel** class represents the artificial intelligence model responsible for generating designs.

**Attributes**

- modelName : String – Name of the AI model
- version : String – Version of the model

**Methods**

- analyzeRoom() – Analyzes room images and details
- generateDesign() – Generates interior design layouts
- fineTuneModel() – Improves the model performance

**Relationship**

- The AI model **processes design requests** and generates design results.

### 7. DesignResult Class

The **DesignResult** class stores the output generated by the AI system.

**Attributes**

- designID : int – Unique design ID
- generatedImage : Image – AI-generated design image
- confidenceScore : float – Accuracy score of the design

**Methods**

- saveDesign() – Saves the generated design
- deleteDesign() – Deletes a design

**Relationship**

- A design request **produces multiple design results**.

### 8. Feedback Class

The **Feedback** class collects user feedback about the generated designs.

**Attributes**

- feedbackID : int – Unique feedback ID
- rating : int – Rating given by the user
- comments : String – User comments

**Method**

- submitFeedback() – Submits feedback to the system.

**Relationship**

- Users can **provide feedback on generated design results**.

**9. Database Class**

The **Database** class is responsible for storing and retrieving system data.

**Methods**

- storeUser() – Stores user data
- storeRoom() – Stores room information
- storePreferences() – Stores user preferences
- storeDesigns() – Stores generated designs
- fetchData() – Retrieves required data

**Relationship**

- The database **stores all system information**, including users, rooms, preferences, and design results.

## 4.4 State Diagram

A **State Diagram** describes the different states of a system and how the system changes from one state to another based on user actions or system events. In the **AI-Driven Interior Design System (Smart Space)**, the state diagram shows how the system moves through different stages from user login to the generation of interior design recommendations.

The system starts when the user opens the application and enters the login credentials. After successful authentication, the user can upload room images and enter room details. The system then allows the user to set design preferences such as style, color scheme, or furniture type. Once the input information is collected.

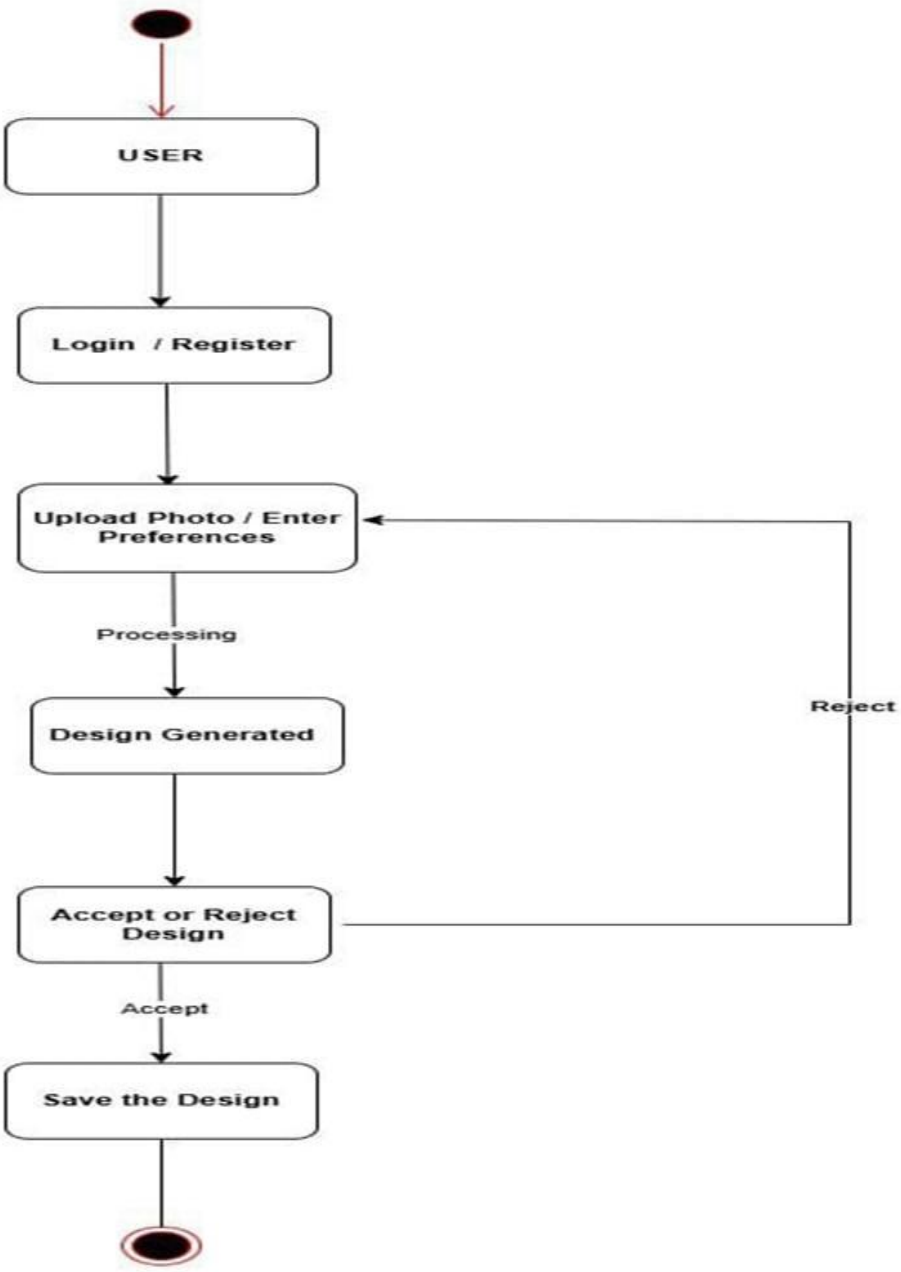



figure 4.4 State Diagram

1. **Start State**

This is the initial state where the user opens the Smart Space application. The system waits for the user to log in.

2. **Loginstate**

In this state, the user enters login credentials such as username and password. The system verifies the information and grants access if the credentials are valid.

3. **Upload Room Image State**

After login, the user uploads a photo of the room that needs interior design suggestions. The system stores and prepares the image for processing.

4. **Enter Room Details State**

The user provides additional information such as room dimensions and type of room. This information helps the AI model understand the room structure.

5. **Set Preferences State**

In this state, the user selects design preferences such as interior style, colors, and furniture type.

6. **AI Processing State**

The system sends all collected data to the AI model. The model analyzes the room image, room details, and user preferences to generate suitable design layouts.

7. **Generate Design State**

The AI system produces interior design images or layouts based on the analysis.

8. **View Design Result State**

The generated interior designs are displayed to the user. The user can review and explore different design suggestions.

## 4.5 Activity Diagram

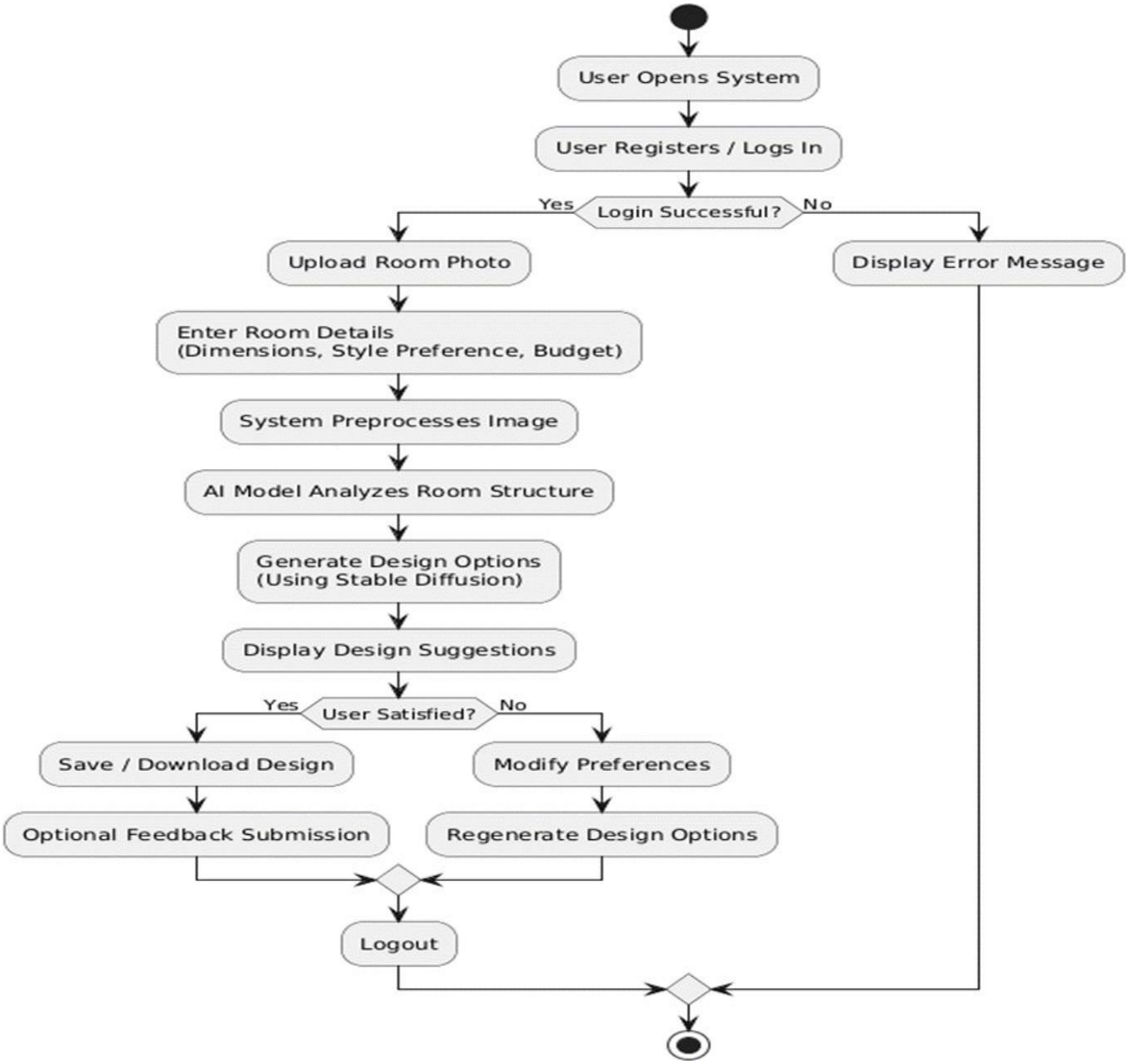



Figure 4.5 Activity Diagram

An **Activity Diagram** represents the workflow of a system by showing the sequence of activities performed to complete a task. It illustrates how different operations are executed step by step from the start of the process to the final result. In the **AI-Driven Interior Design System (Smart Space)**, the activity diagram explains the process of generating interior design recommendations based on user inputs and AI processing.

The workflow begins when the user accesses the Smart Space system and logs into the application. After successful authentication, the user uploads a room image and enters the required details such as room dimensions and type. The user then selects interior design preferences such as style, color scheme, and furniture type. Once all the required inputs are provided, the system sends the data to the AI model.

The AI model analyzes the room image, dimensions, and user preferences to generate appropriate design suggestions. The system processes the data and creates new interior design layouts based on the analysis. After the design generation process is completed, the results are displayed to the user.

The user can then view the generated interior design images and evaluate the suggestions provided by the system. If the user is satisfied, the design can be downloaded or saved. The user may also provide feedback such as ratings or comments about the generated designs. This feedback can help improve the system performance and recommendation accuracy in the future.

The activity diagram clearly shows the sequence of actions performed in the system, starting from user login to viewing and saving the final interior design results. It helps in understanding the overall workflow and operational logic of the AI-driven interior design system.

## 4.6 Sequence Diagram

A **Sequence Diagram** illustrates how different components of a system interact with each other in a specific sequence of time. It shows how messages are exchanged between the user, system modules, and the AI model to complete the process of generating interior design recommendations.

In the **AI-Driven Interior Design System (Smart Space)**, the sequence diagram explains the step-by-step communication between the **User, System Interface, Database, and AI Model** during the interior design generation process.

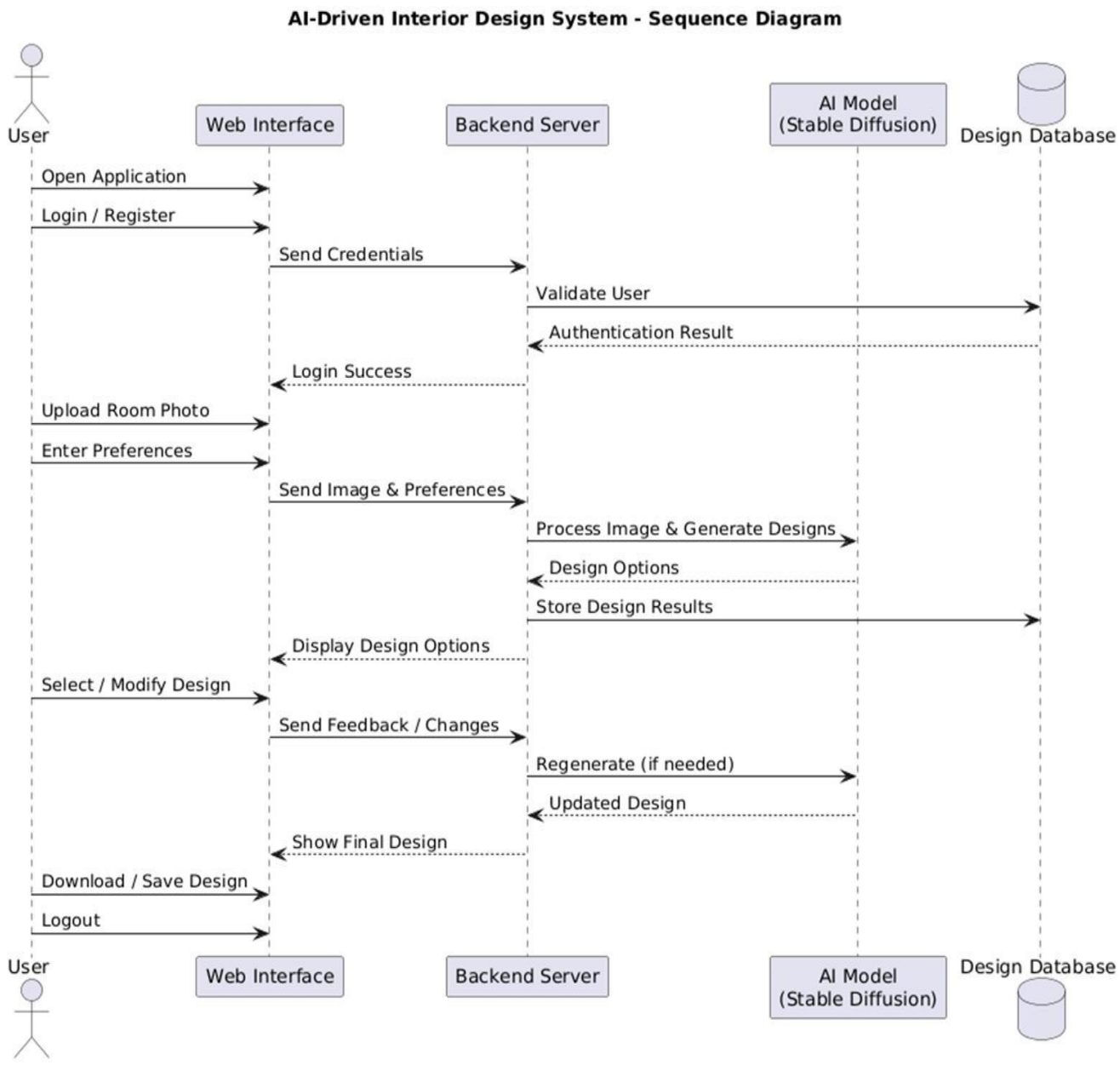



Figure 4.6 Sequence Diagram

**Step 1: Open the Application**

The process begins when the **user opens the Smart Space web application** through the web interface.

**Step 2: Enter Login or Registration Details**

The **user enters login or registration information** such as username and password through the web interface.

**Step 3: Send Credentials to Server**

The **web interface sends the entered credentials to the backend server** for verification.

**Step 4: User Validation**

The **backend server communicates with the design database** to check whether the user exists and whether the password is correct.

**Step 5: Authentication Result**

After verification, the **backend server sends the authentication result back to the web interface**. If the credentials are correct, the user receives a **login success message** and gains access to the system.

**Step 6: Upload Room Image and Preferences**

After login, the user uploads a room photo and enters interior design preferences**,** such as style, color scheme, or furniture type.

**Step 7: Send Data to Backend Server**

The web interface collects the room image and preferences and sends them to the backend server.

**Step 8: AI Processing Request**

The backend server forwards the image and user preferences to the AI model (Stable Diffusion) to generate interior design suggestions.

**Step 9: AI Generates Design Options**

The AI model analyzes the input image and preferences and generates multiple design options.

**Step 10: Return Generated Designs**
The AI model sends the generated design results back to the backend server**.**

**Step 11: Store Results in Database**
The backend server saves the generated designs in the database so that the designs can be accessed later.

**Step 12: Display Designs to User**
The backend server sends the generated designs to the web interface**,** where they are displayed to the user.

**Step 13: User Reviews the Designs**
The **user reviews the generated interior design options** displayed on the interface.

**Step 14: Select or Modify Design**
The user may select a preferred design or request modifications for better results.

**Step 15: Send Feedback to Server**
The web interface sends the user's feedback or modification request to the backend server**.**

**Step 16: Regenerate Design**
If modifications are required, the backend server sends the updated request to the AI model**, which** generates a new design version.

**Step 17: Updated Design Display**
The newly generated design is returned to the backend server and displayed again on the web interface.

**Step 18: Save or Download Design**
Once the user is satisfied with the design, the **user downloads or** saves the final design from the system.

**Step 19: Logout**
Finally, the user logs out of the application**,** and the session ends.

# CHAPTER 5
# METHODOLOGY

## 5.1 module Description

The methodology describes the systematic approach used to design and implement the **AI-Driven Interior Design System (Smart Space)**. This project follows a structured process that integrates user interaction, data processing, and artificial intelligence techniques to generate interior design recommendations. The methodology ensures that the system efficiently collects user inputs, processes them using AI models, and produces visually appealing interior design suggestions.

The development process begins with **data collection and user input**. The user interacts with the web interface by uploading room images and providing information such as room dimensions and design preferences. These preferences may include interior design style, color scheme, and furniture type. The system collects this data and prepares it for further processing.

The next step involves **data preprocessing and validation**. The uploaded room images are checked for format and quality, and the room details are verified to ensure accuracy. This preprocessing step helps the system organize the input data and makes it suitable for AI model processing. At the same time, user credentials and preferences are stored in the database for secure and efficient access.

After preprocessing, the system proceeds to the **AI-based design generation phase**. The processed room images and user preferences are sent to the AI model, which uses advanced machine learning techniques to analyze the room structure and generate suitable interior design layouts. In this project, an AI image generation model such as **Stable Diffusion** can be used to transform the input image into a redesigned interior space according to the selected style.

Once the AI model generates the design outputs, the system performs **result processing and storage**. The generated interior design images are stored in the database and organized for easy retrieval. This ensures that users can view their previously generated designs without losing the results.

The system then moves to the **user interaction and visualization phase**. The generated designs are displayed to the user through the web interface, allowing the user to view different design options. The user can evaluate these results and select the most suitable design based on personal preferences.

Another important part of the methodology is **feedback and refinement**. If the user is not satisfied with the generated design, the system allows the user to modify preferences or request additional design variations. The updated inputs are again processed by the AI model to generate improved design suggestions.

Finally, the system reaches the **finalization stage**. In this stage, the user can download or save the selected design for future use. The system also allows users to provide feedback, which can help improve the accuracy and performance of the AI model in future implementations.

Overall, the methodology ensures a structured workflow that integrates user interaction, AI-based image generation, and database management to provide intelligent interior design recommendations efficiently

### 5.1.2 Data Collection and Preprocessing Module

The **Data Collection and Preprocessing Module** is the first and one of the most important stages of the AI-Driven Interior Design System. This module is responsible for gathering the necessary input data from the user and preparing it for further processing by the AI model. Proper data collection and preprocessing ensure that the system receives accurate and well-structured information, which improves the quality of the generated interior design results.

The process begins with **data collection from the user**. The user interacts with the system through the web interface and uploads a room image that needs interior design suggestions. Along with the image, the user may also provide additional information such as room dimensions, room type (bedroom, living room, kitchen, etc.), and design preferences. These preferences may include interior design styles such as modern, minimalist, or traditional, as well as color schemes and

furniture types. All this information is collected and stored temporarily in the system for further processing.

Once the data is collected, the system performs data validation to ensure that the input information is correct and usable. The system checks whether the uploaded image is in a supported format such as JPEG or PNG and verifies that the file size is within the allowed limit. Similarly, the system verifies that the room dimensions and other details entered by the user are valid.

After validation, the system moves to the data preprocessing stage**.** In this step, the uploaded room image is prepared for AI processing. The image may be resized, normalized, or converted into a format that is suitable for the AI model. Preprocessing may also involve removing unnecessary noise from the image and adjusting brightness or contrast to improve image clarity. These steps help the AI model better analyze the room environment.

At the same time, the user-provided text data such as design preferences and room details are structured and organized in a format that the system can easily process. This structured data is then stored in the database along with the image information.

Finally, the preprocessed image and structured user data are forwarded to the next module, which is the **AI Design Generation Module**. By ensuring that the input data is clean, valid, and properly formatted, the Data Collection and Preprocessing Module plays a critical role in improving the accuracy and efficiency of the AI-driven interior design system.

### 5.1.3 ControlNet and Stable Diffusion Pipeline Module

The **ControlNet and Stable Diffusion Pipeline Module** is the core component of the AI-Driven Interior Design System. This module is responsible for generating new interior design images based on the user's room image and design preferences. It uses advanced artificial intelligence techniques to transform the existing room layout into a redesigned interior space while maintaining the original structure of the room. techniques to transform the existing room layout into a redesigned interior space while maintaining the original structure of the room.

The process begins when the **preprocessed room image and user preferences** are received from the previous module. The system first sends the room image to the **ControlNet model**, which helps the AI understand the structural details of the room. ControlNet analyzes the image and extracts important features such as room layout, walls, furniture positions, edges, and depth information. This step ensures that the generated design maintains the original spatial arrangement of the room.

After extracting the structural information, the processed data is passed to the Stable Diffusion mode**l**. Stable Diffusion is a powerful image generation model that creates new images based on textual prompts and visual input. In this system, the prompt may include the user's design preferences such as interior style, color theme, lighting style, or furniture arrangement. The Stable Diffusion model uses this information to generate realistic interior design images.

The pipeline process combines both ControlNet and Stable Diffusion to produce accurate and visually appealing results. ControlNet ensures that the generated image follows the original room structure, while Stable Diffusion focuses on creating creative and high-quality interior design variations. This combination allows the system to produce multiple design options for the same room.

During the generation process, the AI model may produce several design outputs with different layouts and styles. These outputs are then refined and filtered to ensure clarity and quality. The system selects the most suitable design images and prepares them for display.

Finally, the generated interior design images are sent to the Result Processing Module**,** where they are stored in the database and displayed to the user through the application interface. This module plays a critical role in converting user input and room images into realistic and innovative interior design suggestions using advanced AI techniques. module plays a critical role in converting user input and room images into realistic and innovative interior design suggestions using advanced AI techniques.

### 5.1.4 Fine-Tuning Module

The Fine-Tuning Module is an important component of the AI-Driven Interior Design System. This module focuses on improving the performance and accuracy of the AI model so that it can generate better interior design results. Fine-tuning helps the system adapt the general AI model to the specific requirements of interior design applications.

The process begins with the collection of training data. The system uses a dataset that contains different types of room images along with corresponding interior design styles. These images may include various room types such as bedrooms, living rooms, kitchens, and office spaces. Each image is associated with design styles, color themes, and furniture arrangements. This dataset helps the AI model learn the relationship between room structures and design patterns.

After collecting the dataset, the **Stable Diffusion model is fine-tuned** using this data. Fine-tuning means adjusting the parameters of the pre-trained model so that it becomes more specialized for interior design tasks. During this process, the model learns to recognize room layouts, furniture placement, lighting conditions, and decoration styles. This improves the model's ability to generate realistic and relevant interior design images.

The fine-tuning process also involves training the model with design prompts and image examples. For example, prompts such as "modern living room with minimal furniture" or "luxury bedroom with warm lighting" help the model understand how different styles should appear visually. By learning from these examples, the AI model can produce more accurate and creative designs based on user input.

Another important function of the Fine-Tuning Module is performance optimization. The system evaluates the generated designs and compares them with expected design outcomes. Based on the evaluation results, the model parameters are adjusted to reduce errors and improve image quality. This step ensures that the generated interior designs look realistic and match the user's preferences.

Finally, the improved AI model is integrated back into the ControlNet and Stable Diffusion pipeline used in the system. The fine-tuned model is then capable of generating more accurate and

visually appealing interior design suggestions. As a result, the Fine-Tuning Module enhances the overall efficiency and reliability of the AI-Driven Interior Design System.

**5.1.5**

**Inference Module**

The **Inference Module** is responsible for generating the final interior design results using the trained AI model. In this stage, the system uses the **fine-tuned Stable Diffusion and ControlNet model** to produce interior design images based on the user's input. Unlike the training phase, the inference stage focuses only on generating results without modifying the model parameters.

The process begins when the **preprocessed room image and user preferences** are received from the previous modules. These inputs may include the uploaded room image, room dimensions, and design preferences such as interior style, color theme, and furniture arrangement. The system prepares this information and sends it to the AI model for processing.

The **ControlNet model** first analyzes the structure of the uploaded room image. It extracts important visual features such as edges, layout, and spatial arrangement of objects in the room. This ensures that the generated design maintains the original structure and proportions of the room.

After extracting the structural information, the data is passed to the **Stable Diffusion model**. The Stable Diffusion model uses the input image along with the textual prompt representing user preferences to generate new interior design images. The model gradually transforms the input image through multiple steps of the diffusion process until a new design layout is created.

During the inference stage, the system may generate **multiple design variations** for the same room. These variations allow users to explore different design possibilities based on their selected style. The generated images are then evaluated and filtered to ensure that they meet the required quality and clarity standards.

Finally, the generated interior design images are sent to the **output module**, where they are stored in the database and displayed to the user through the web interface. The user can then view, download, or provide feedback on the generated designs.

Overall, the **Inference Module plays a crucial role in transforming user input into visually appealing interior design recommendations** using the trained AI model, making the system practical and useful for real-world applications.

## 5.2 Home page

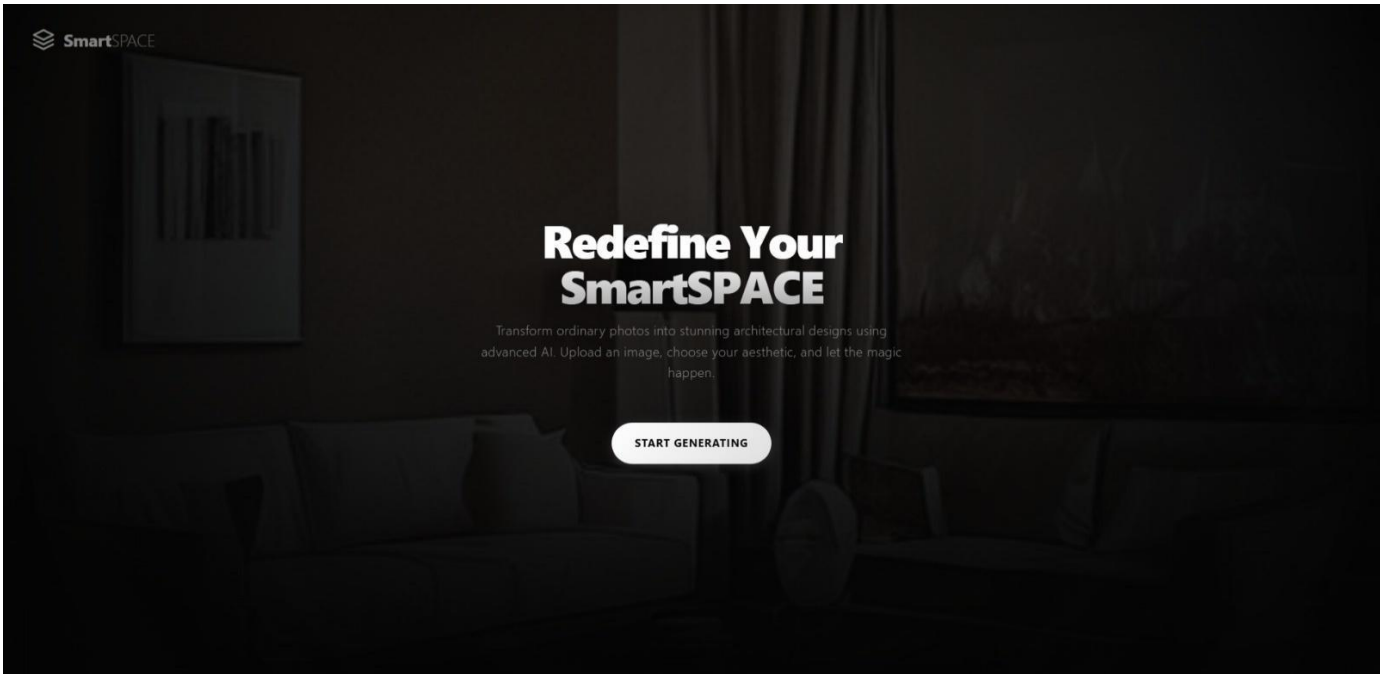
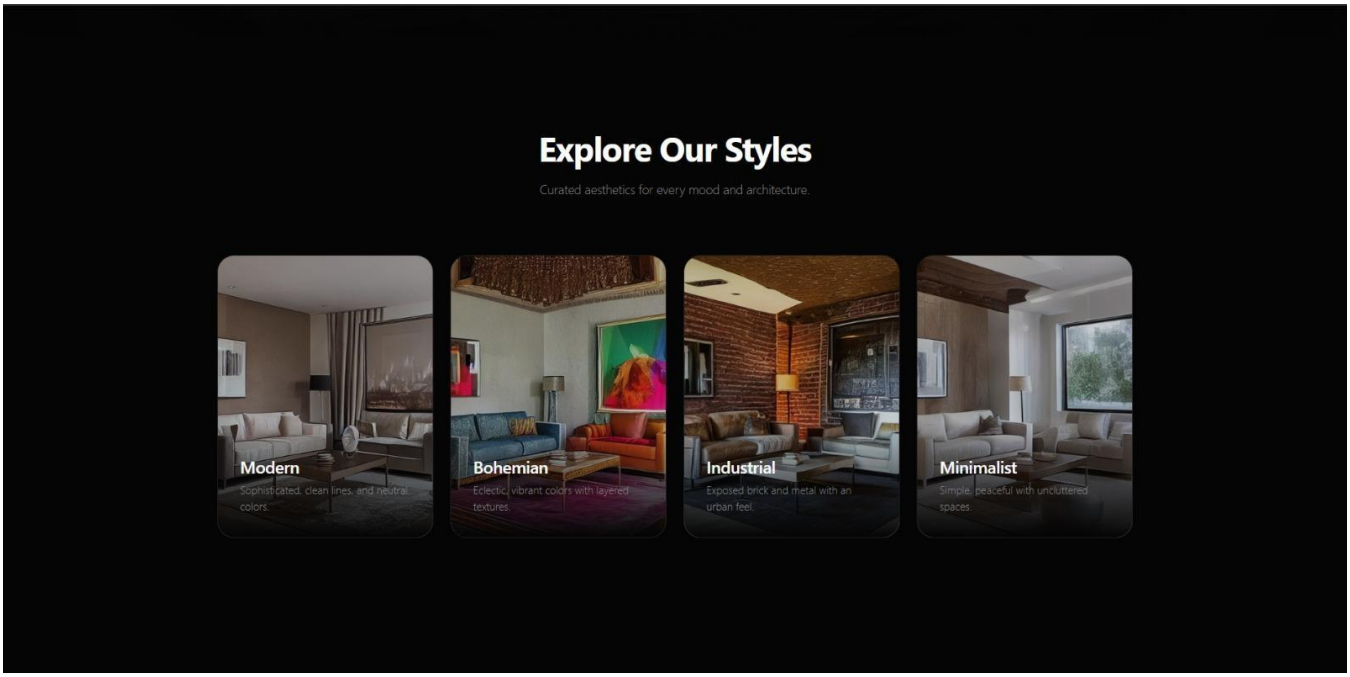



Figure 5.2 Home page

The user opens the SmartSPACE application landing page. They click the start generating button to begin the process. The user then enters their login or registration details to access the system. The web interface sends these credentials to the backend server for verification. The server validates the user through the design database. Once authenticated, the user uploads a photo of their room and enters their style preferences. The web interface sends the image and the preferences to the backend server. The server passes this information to the Stable Diffusion AI model to generate designs. The AI model creates design options and sends them back to the server. The server stores these results in the design database and displays them to the user. The user

reviews the designs to select or modify an option. The server requests the AI to update the design based on user feedback. Finally, the user saves the finished design and logs out of the application.

. The server passes this information to the Stable Diffusion AI model to generate designs. The AI model creates design options and sends them back to the server.




The user scrolls to the "Explore Our Styles" section of the website. This page presents four distinct interior design aesthetics for the user to consider. The Modern style offers sophisticated lines and neutral colors for a clean look. The Bohemian option features eclectic and vibrant colors with layered textures. The Industrial choice focuses on exposed brick and metal for an urban feel. Finally, the Minimalist style provides a simple and peaceful environment with uncluttered spaces. This visual guide helps the user decide on their preferences before starting the AI generation process. The user navigates to the style selection section of the SmartSPACE platform. This page displays a variety of curated aesthetics for different architectural moods. The user views four primary categories including Modern, Bohemian, Industrial, and Minimalist. Each category provides a visual example and a brief description of the design characteristics. The user interacts with these options to determine the look they want for their room. This step directly supports the

preference entry phase of the design workflow. By selecting a style, the user provides the AI with specific artistic constraints. The system uses these choices to guide the Stable Diffusion model during the image generation process. This visual gallery ensures the user knows exactly what kind of output to expect from the system.

### 5.2.1 Main Page

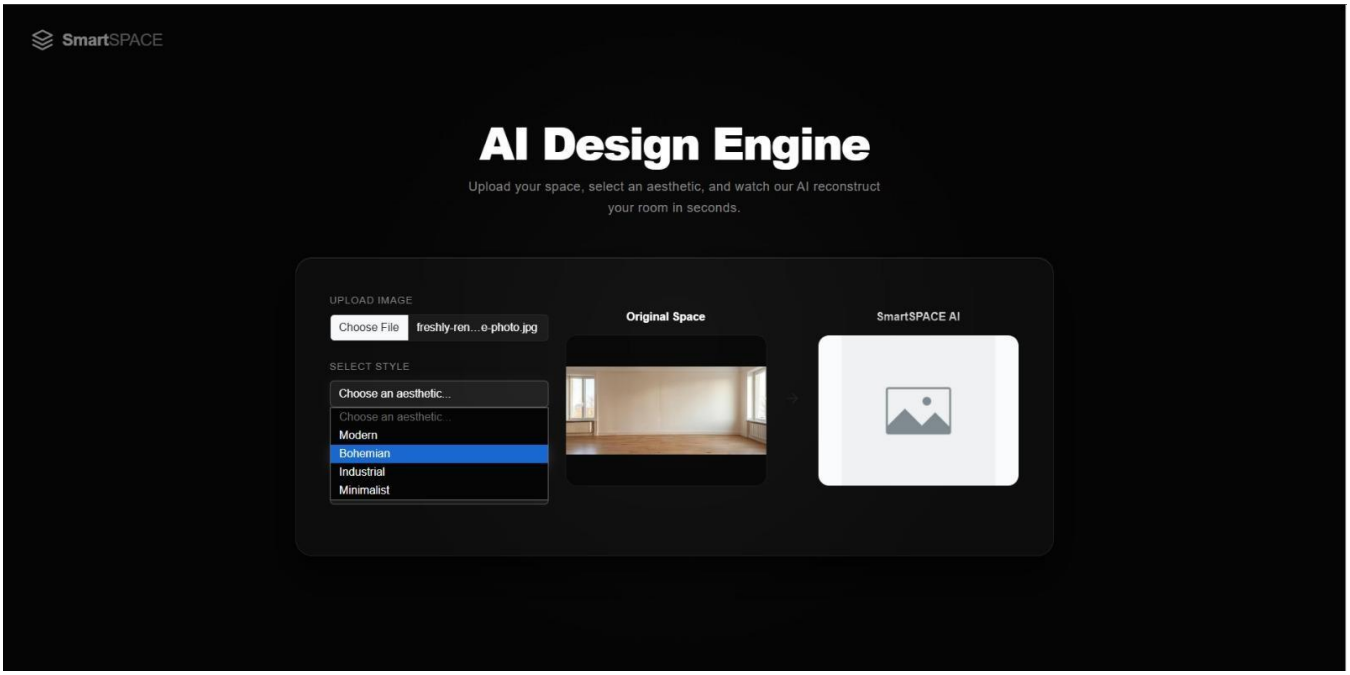
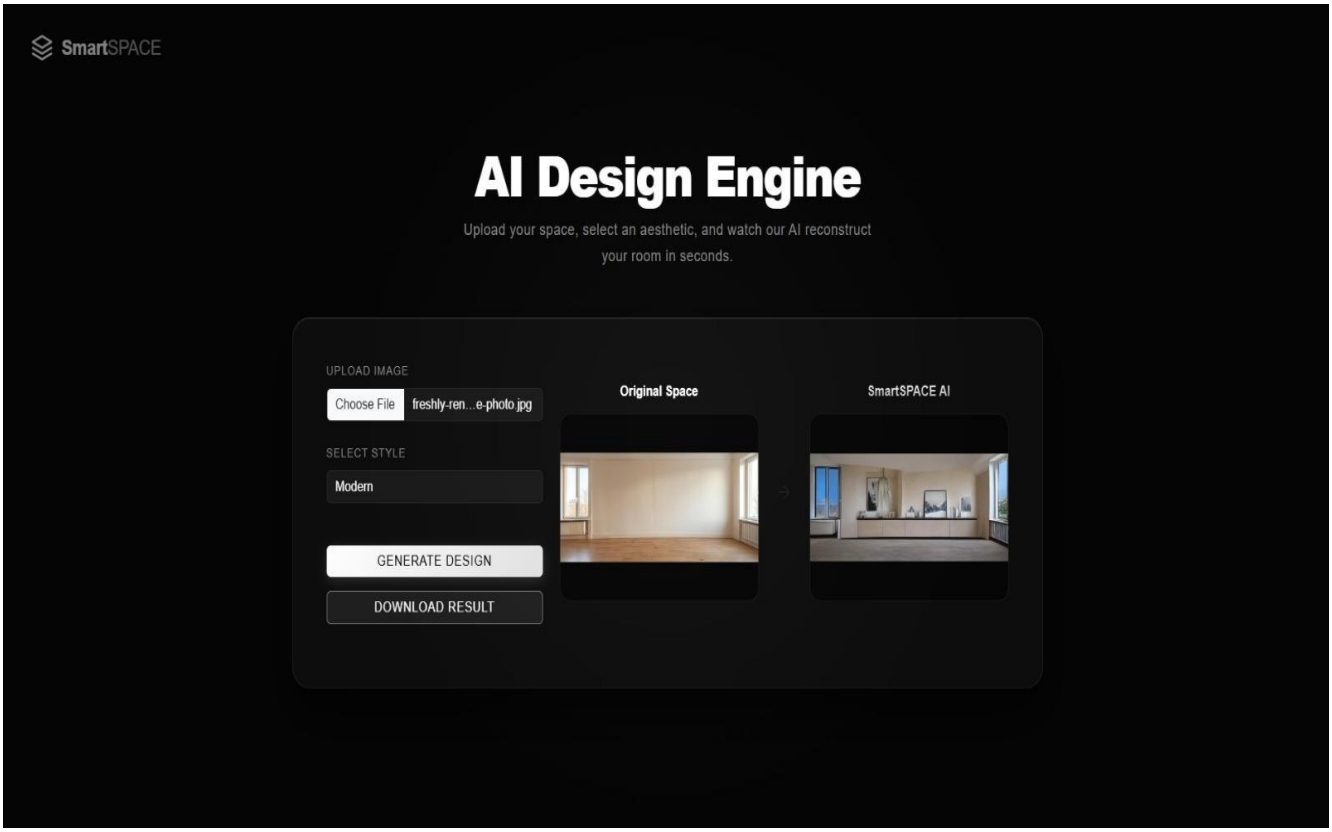



Figure 5.2.1 main page

The user interacts with the AI Design Engine to transform their living area. They start by selecting the "Choose File" button to upload a high-quality photo of their current room. The interface displays a thumbnail of this image under the "Original Space" heading for confirmation. The user then moves to the "Select Style" section to define the desired look. They click the dropdown menu which reveals options like Modern, Bohemian, Industrial, and Minimalist. For this specific session, the user highlights the Bohemian option to apply a vibrant and eclectic theme. A placeholder icon appears in the "SmartSPACE AI" box to indicate where the result will be shown. This dashboard serves as the primary control center for the user's input before the backend server takes over. By providing these specific details, the user ensures the AI has the necessary data to perform an

input before the backend server takes over. By providing these specific details, the user ensures the AI has the necessary data to perform .

accurate reconstruction. The design is now ready to be processed by the Stable Diffusion model in the next step of the sequence.




The user triggers the design process by clicking the generate design button. The backend server receives the room photo and the selected Modern aesthetic for processing. The AI model analyzes the layout of the original space to maintain its structural integrity. It then applies the new textures, furniture, and lighting according to the chosen style. The generated design appears in the SmartSPACE AI preview window for the user to review. This result shows a clean and sophisticated transformation of the empty room. The user now has the option to click the download result button to save the image. This completion step fulfills the final actions shown in your

sequence diagram. The system has successfully moved from data input to a finished architectural reconstruction.

### 5.3 System Architecture

The **System Architecture** describes the overall structure of the AI-Driven Interior Design System and explains how different components interact with each other to generate interior design suggestions. It shows how data flows from the user interface to the backend system, how the AI model processes the data, and how the results are delivered to the user.

The architecture of the system is designed using a **multi-layer structure**, which includes the **User Interface Layer, Application/Backend Layer, AI Processing Layer, and Database Layer**. Each layer performs a specific function to ensure efficient operation of the system.

The first layer is the **User Interface Layer**. This layer is responsible for interacting with the user. Through the web interface, the user can register or log in to the system, upload room images, enter room details, and select interior design preferences such as style and color scheme. The interface collects these inputs and sends them to the backend server for processing. This layer ensures a user-friendly interaction between the user and the system.

The second layer is the **Application or Backend Layer**. This layer manages the core system operations and controls the workflow of the application. The backend server receives user inputs from the web interface and performs tasks such as authentication, data validation, and request management. It organizes the input data and prepares it for AI processing. The backend also communicates with the database to store and retrieve user information, room details, and generated design results.

The third layer is the **AI Processing Layer**. This layer is responsible for generating interior design suggestions using artificial intelligence models. In this project, models such as **ControlNet and Stable Diffusion** are used. ControlNet analyzes the structure of the uploaded room image and extracts layout information such as edges and spatial arrangement. Stable Diffusion then uses this information along with the user’s design preferences to generate realistic interior design images. This layer acts as the intelligence of the system.The final layer is the **Database Layer**. This layer

stores all the system data including user accounts, room images, room dimensions, design preferences, and generated design results. The database ensures that data can be retrieved whenever required and helps maintain the history of generated designs for each user.The data flow in the system begins when the user uploads a room image and enters preferences through the interface. The backend server receives and processes this information, then sends it to the AI model for design generation. The AI model generates new interior design images and sends the results back to the backend. Finally, the results are stored in the database and displayed to the user through the web interface.

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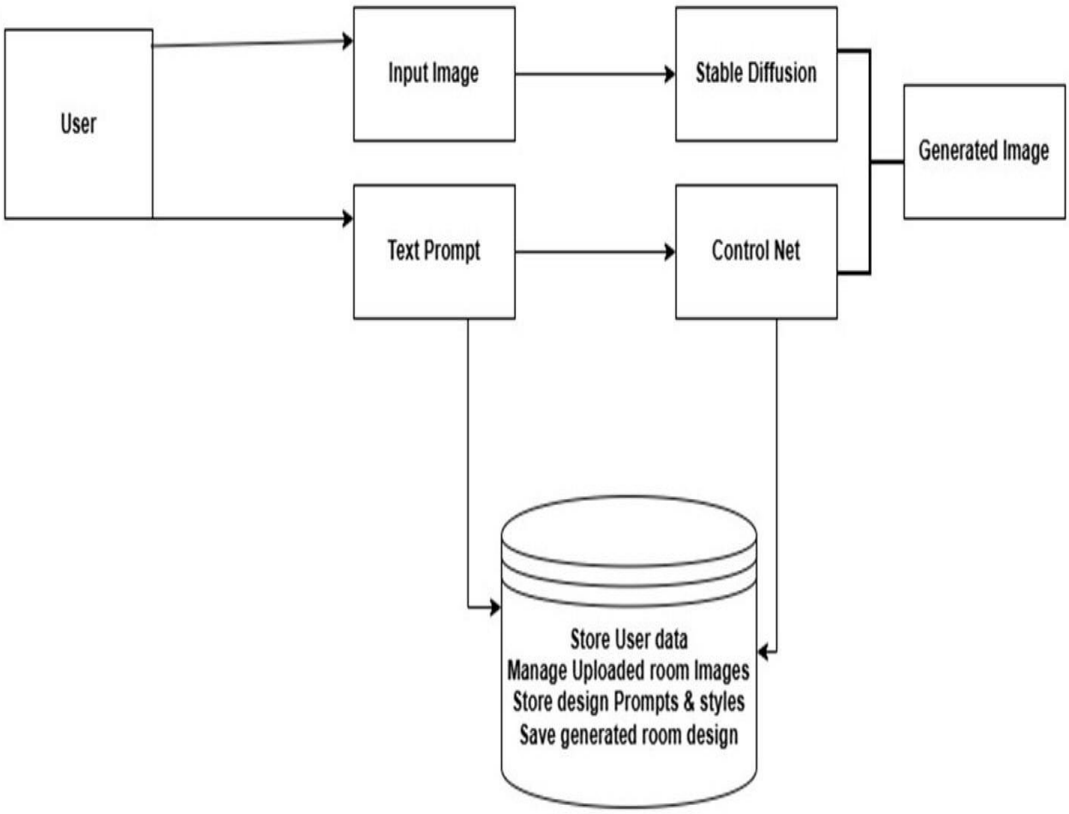


Figure 5.3 System Architecture

The **system architecture** of the AI-driven interior design system explains how different components work together to transform a user's room photo into a newly generated interior design. The entire process begins with the **user**, who acts as the primary source of information and creative direction for the system. The user interacts with the application through the web interface and

provides two important types of input. The first input is the **room image**, which is a photograph of the actual room that the user wants to redesign. This image gives the system visual information

about the room structure, including the layout, walls, floor space, furniture placement, windows, and lighting conditions. The second input is the **text prompt or design preference**, which describes the style that the user wants the room to follow. These prompts may include design styles such as modern, minimalist, traditional, rustic, luxury, or bohemian. The combination of the room image and the design prompt provides the foundation for the system to generate customized interior designs.

Once the user uploads the image and enters the design prompt, both types of data are transferred to the **backend server** where the system begins processing the request. At this stage, the information is temporarily stored in the **database**, which acts as a central storage component of the system. The database keeps records of the uploaded images, user preferences, and generated results. Storing this information ensures that the user can later retrieve previously generated designs, download them, or modify them if necessary. The database also helps maintain a structured dataset that can be used for future system improvements.

After the initial storage process, the input data moves to the **AI processing layer**, which is the most important part of the system architecture. In this layer, two main AI technologies work together: **Stable Diffusion** and **ControlNet**. These two models cooperate to produce high-quality interior design images while preserving the structure of the original room. The room image is first analyzed by the Stable Diffusion model, which is responsible for generating new visual content. Stable Diffusion is a powerful generative AI model that can create realistic images by gradually refining visual details through a diffusion process. It uses the information from the room image and the design prompt to understand the user's requirements.

At the same time, the **ControlNet module** plays a guiding role in the generation process. ControlNet examines the structural details of the room image, such as edges, layout lines, and object boundaries. This information is used to ensure that the generated design keeps the same physical structure as the original room. For example, the positions of walls, doors, windows, and

floors remain consistent even when the interior decoration changes. Without ControlNet, the AI might generate unrealistic designs where walls move or objects appear in incorrect locations. By guiding the Stable Diffusion model, ControlNet ensures that the generated interior design looks both creative and structurally accurate.The **Stable Diffusion model and ControlNet module operate together in a coordinated pipeline**. The diffusion model focuses on generating new textures, furniture arrangements, lighting styles, and decorative elements according to the text prompt. Meanwhile, ControlNet continuously provides structural guidance so that the generated design fits naturally into the original room layout. Through this collaboration, the system gradually transforms the original room photo into a redesigned interior space that matches the user's chosen style.

During this generation stage, the system may produce **multiple design variations** for the same room. These variations allow the user to compare different interior design possibilities. For example, the same room may be generated in modern, minimalist, or luxury styles depending on the prompt. The system evaluates the generated images to ensure that they meet quality standards and removes results that appear unclear or unrealistic.After the generation process is completed, the final **generated interior design images** are sent back to the backend server. The system then stores these generated designs in the database along with the original room image and user prompt. This storage step is important because it allows the user to access the generated designs later without needing to repeat the generation process.Finally, the **generated room designs are delivered back to the user interface**, where the user can view them in the application. The user can explore different design options, select the most suitable design, download the generated image, or request further modifications. If the user wants to refine the design, they can change the text prompt or style preference, and the system will generate a new set of design variations.Overall, the system architecture creates a seamless workflow that integrates **user interaction, database management, and advanced AI image generation models**. By combining Stable Diffusion with ControlNet, the system ensures that the generated interior designs are both visually appealing and structurally consistent with the original room. This architecture allows users to easily transform their real rooms into creative interior designs using the power of artificial intelligence.

# CHAPTER 6

# IMPLEMENTATION DETAILS

The **implementation phase** describes how the AI-Driven Interior Design System is developed and integrated using different technologies, tools, and frameworks. This phase focuses on converting the system design and architecture into a working application that can accept user input, process the data using artificial intelligence models, and generate interior design suggestions.

The system is implemented using a **web-based architecture** where the user interacts with the application through a browser interface. The front-end interface is designed to provide an easy and user-friendly environment where users can upload room images, enter design preferences, and view the generated results. Technologies such as **HTML, CSS, and JavaScript** are commonly used to design the front-end interface, ensuring smooth interaction between the user and the system.

The **backend system** is responsible for handling user requests, managing data, and coordinating communication between the user interface and the AI model. A backend framework such as **Python-based frameworks (Flask or Django)** can be used to implement the server-side logic. The backend processes user inputs, validates the uploaded images, and prepares the data for AI model processing. It also manages authentication features such as login and registration to ensure that only authorized users can access the system.The **database system** is implemented to store all relevant information required by the application. The database stores user account details, uploaded room images, user design preferences, and generated interior design results. A database management system such as **MySQL or MongoDB** can be used for this purpose. The database ensures that the data is stored securely and can be retrieved whenever the user wants to access previously generated designs.The most important component of the implementation is the **AI processing module**, which uses models such as **Stable Diffusion and ControlNet**. These models are integrated into the system using Python libraries such as **PyTorch and Diffusers**. The Stable Diffusion model is responsible for generating new interior design images based on the input room image and the user's design prompt. ControlNet is used to guide the generation process by

preserving the structural layout of the original room. This integration ensures that the generated designs maintain the same room structure while applying new styles and decorations.

During implementation, the system also includes **image preprocessing techniques**. Before sending the image to the AI model, the system performs operations such as resizing, normalization, and format conversion. These preprocessing steps improve the performance and accuracy of the AI model.

The generated designs are processed and returned to the backend server, where they are stored in the database and displayed to the user through the web interface. The system allows the user to view multiple design options, download the preferred design, or generate new variations by modifying the design prompt.

In addition, the implementation includes **error handling and performance optimization**. The system checks for invalid inputs such as unsupported image formats or incorrect login credentials. Efficient data handling and optimized model execution are used to reduce processing time and improve system responsiveness

## 6.1 Testing Methods

### 6.1.1 Unit Testing

**Unit testing** is the process of testing individual components or modules of the system separately to ensure that each part functions correctly. In this method, every small part of the application is tested independently before combining it with other modules. The main objective of unit testing is to identify errors in the early stage of development.In the AI-Driven Interior Design System, several modules are tested individually. For example, the **login module** is tested to check whether the system correctly verifies user credentials. The **image upload module** is tested to ensure that users can successfully upload room images without errors. The **data preprocessing module** is also tested to verify that the system correctly resizes and prepares images before sending them to the AI model. By testing each module separately, developers can easily detect and fix problems before integrating them into the full system.

### 6.1.2 Integration Testing

**Integration testing** is performed after unit testing. In this method, different modules of the system are combined and tested together to verify that they communicate and interact properly. The purpose of integration testing is to ensure that data flows correctly between different components of the system.

In the AI-driven interior design system, integration testing checks the connection between the **user interface, backend server, AI model, and database**. For example, the system verifies whether the room image uploaded from the web interface is successfully transferred to the backend server. It also checks whether the backend server correctly sends the image and user preferences to the **Stable Diffusion and ControlNet AI models**. Another important test is verifying whether the generated interior design images are properly stored in the database and displayed back to the user through the interface. This type of testing ensures that all modules work together smoothly.

### 6.1.3. System Testing

**System testing** is the process of testing the complete and fully integrated system to ensure that it meets the required functionality and performance standards. Unlike unit testing and integration testing, which focus on specific modules, system testing evaluates the entire application as a whole.

In the AI-driven interior design system, system testing involves checking the full workflow of the application. The tester verifies whether a user can successfully register or log in, upload a room image, enter design preferences, and receive AI-generated interior design suggestions. The system also checks whether the generated images are realistic and whether the user can download or save the design results. Performance and reliability of the system are also tested to ensure that the application works efficiently under normal usage conditions. and whether the user can download or save the design results. Performance and reliability of the system are also tested to ensure that the application works efficiently under normal usage conditions.

# CHAPTER 7

# ADVANTAGES AND DISADVANTAGES

## 7.1 Advantages

### 1. Time Saving

The system helps users generate interior design ideas quickly without spending a long time planning or consulting professional designers.

### 2. Multiple Design Options

The AI model can generate several interior design variations for the same room, allowing users to explore different styles and layouts.

### 3. Cost Effective

Users can visualize different interior design concepts without hiring expensive professional interior designers.

### 4. Automated Design Process

The system automatically analyzes room images and generates design layouts using artificial intelligence, reducing manual effort.

### 5. Easy Accessibility

The system can be accessed through a web interface, allowing users to upload images and generate designs from anywhere.

### 6. User Friendly Interface

The application is designed with a simple interface so that users can easily upload images, select preferences, and view generated designs.

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### 7. Creative Design Suggestions

The AI model can produce creative interior design ideas that users might not have considered before.

### 8. Quick Visualization

Users can quickly visualize how their room might look with different interior styles before making real changes.

### 9. Digital Storage of Designs

Generated designs are stored in the database, allowing users to access or download them later.

### 10. Improved Decision Making

By comparing multiple design options, users can make better decisions about how to decorate or renovate their rooms.

## 7.2 Disadvantages

### 1. Accuracy Limitations

The quality of generated designs depends on the clarity of the input image and the quality of user prompts.

### 2. High Computational Requirements

AI models such as Stable Diffusion require powerful hardware resources like GPUs and high memory.

### 3. Slow Processing Time

Generating high-quality interior design images may take time depending on system performance.

### 4. Dependence on Internet Connectivity

If the system is web-based, users need a stable internet connection to upload images and receive results.

### 5. Limited Practical Considerations

The system focuses mainly on visual design and may not consider practical aspects like construction constraints or material availability.

### 6. Limited Training Data

If the AI model is trained with limited datasets, the variety of design outputs may be restricted.

### 7. Not a Complete Replacement for Designers

The system provides design ideas but cannot fully replace professional interior designers who consider real-world factors.

### 8. Possible Design Errors

Sometimes the AI may generate unrealistic furniture placement or design elements.

### 9. Security and Privacy Concerns

User-uploaded images and data must be properly protected to avoid privacy issues.

### 10. Maintenance Requirements

The system requires regular updates, model improvements, and maintenance to keep it working efficiently.

# CHAPTER 8
# RESULT AND DISCUSSION

## 8.1 System Overview

### 1. System Performance

The developed AI-Driven Interior Design System was successfully implemented and tested using different room images and design prompts. The system was able to accept user inputs such as room photographs and design preferences through the web interface. After processing the inputs, the AI model generated interior design suggestions within a reasonable time. The system performed efficiently in handling user requests and delivering the generated designs through the interface.

### 2. Design Generation Results

The system produced multiple interior design variations for each uploaded room image. These generated images demonstrated different design styles such as modern, minimalist, luxury, and traditional interiors. The AI model successfully transformed the appearance of the room by modifying elements such as furniture arrangement, wall decoration, lighting style, and color themes. This provided users with a wide range of design possibilities for the same room.

### 3. Structural Consistency of Generated Designs

One of the major strengths of the system was its ability to maintain the structural layout of the original room. By integrating ControlNet with the Stable Diffusion model, the system preserved important structural elements such as walls, doors, and windows while generating new design ideas. This ensured that the generated interior designs remained realistic and suitable for the original room space.

### 4. User Experience and Interaction

The user interface allowed users to easily upload images, select design preferences, and view the generated results. The generated designs were displayed clearly on the screen, allowing users to

compare different options and select the most suitable one. Users were also able to download the generated designs or generate new variations by modifying the design prompt.

**5. Observed Limitations**

During testing, a few limitations were observed in the generated results. In some cases, the AI model produced designs with minor visual inconsistencies such as unrealistic furniture placement or unusual decorative elements. These issues were mostly observed when the input room image was of low quality or when the design prompt provided by the user lacked sufficient detail.

**6. Performance Considerations**

The generation of interior design images required significant computational resources. Systems equipped with graphical processing units (GPUs) were able to produce results faster than systems relying only on central processing units (CPUs). This indicates that hardware performance plays an important role in improving the speed and efficiency of AI-based image generation systems.

**7. Overall Discussion**

The results demonstrate that the AI-driven interior design system is capable of generating creative and visually appealing design suggestions based on user inputs. The integration of Stable Diffusion and ControlNet models improved the realism and structural consistency of the generated images. Although some limitations exist, the system provides an effective tool for users to explore interior design ideas and visualize room transformations before implementing them in real life.

# CHAPTER 9
# CONCLUSION

## 9.1 Summary

The AI-Driven Interior Design System was developed to assist users in generating creative and visually appealing interior design ideas using artificial intelligence. The system allows users to upload a room image and provide design preferences such as style, theme, or furniture arrangement. Using advanced AI models like Stable Diffusion and ControlNet, the system analyzes the room structure and generates multiple interior design variations while maintaining the original layout of the room.The developed system successfully demonstrates how artificial intelligence can simplify the interior design process. It helps users visualize different design possibilities quickly and efficiently. The integration of AI models with a web-based interface allows users to interact with the system easily and obtain design suggestions within a short time. Overall, the system provides a convenient solution for generating interior design concepts and improving user decision-making.

## 9.2 Limitations

Although the system provides useful design suggestions, certain limitations were observed during development and testing.The accuracy of generated designs mainly depends on the quality of the input image and the clarity of the user's design prompt. If the input image is unclear or poorly captured, the generated results may not be fully realistic. Another limitation is the high computational requirement of AI models like Stable Diffusion, which may increase processing time on systems without powerful hardware such as GPUs.

In some cases, the generated designs may contain minor visual inconsistencies, such as unusual furniture placement or unrealistic decorative elements. The system also focuses mainly on visual design generation and does not fully consider real-world factors such as construction limitations, material availability, or cost estimation.

### 9.3 Future Scope

The system can be further improved by integrating more advanced AI models and larger training datasets to increase the accuracy and diversity of generated designs. Future versions of the system could include additional features such as 3D interior design visualization, real-time room editing, and interactive design customization.Another possible improvement is the integration of augmented reality (AR) technology, which would allow users to visualize the generated designs directly in their real rooms through mobile devices. The system could also include cost estimation features and furniture recommendation systems to help users plan their interior design more effectively.Improving system performance and reducing image generation time through optimized algorithms and better hardware support can also enhance the overall user experience

.

### 9.4 Applications

The AI-Driven Interior Design System can be used in several practical applications.It can be used by homeowners who want to redesign or renovate their rooms by exploring different interior design styles. Interior designers can use the system as a supportive tool to quickly generate design ideas and present them to clients.

The system can also be useful for real estate companies to visualize interior design possibilities for empty apartments or houses. Furniture companies can use the technology to recommend furniture arrangements for customers. Additionally, the system can be used in architecture and design education to help students understand different interior design concepts and styles.

Overall, the proposed system has the potential to become a valuable tool in the field of interior design by combining artificial intelligence with creative design visualization. Overall, the proposed system has the potential to become a valuable tool in the field of interior design by combining artificial intelligence with creative design visualization.
Overall, the proposed system has the potential to become a valuable tool in the field of interior design by combining artificial intelligence with creative design visualization.

# CHAPTER 10

# APPENDICES

app > src > Home.jsx > ...

```jsx
import React, { useState, useEffect } from 'react';
import { Container, Row, Col } from 'react-bootstrap';
import { Link } from 'react-router-dom';
import './Home.css';

// Importing images directly
import style1 from './assets/style1.png';
import style2 from './assets/style2.png';
import style3 from './assets/style3.png';
import style4 from './assets/style4.png';

import sample1 from './assets/sample1.png';
import sample2 from './assets/sample2.png';
import sample3 from './assets/sample3.png';
import sample4 from './assets/sample4.png';

const styles = [
  { name: "Modern", description: "Sophisticated, clean lines, and neutral colors.", image: style1 },
  { name: "Bohemian", description: "Eclectic, vibrant colors with layered textures.", image: style2 },
  { name: "Industrial", description: "Exposed brick and metal with an urban feel.", image: style3 },
  { name: "Minimalist", description: "Simple, peaceful with uncluttered spaces.", image: style4 },
];

const heroImages = [style1, style2, style3, style4];
const sampleImages = [sample1, sample2, sample3, sample4];

function Home() {
  const [currentBg, setCurrentBg] = useState(0);

  // Background Slideshow Effect
  useEffect(() => {
    const interval = setInterval(() => {
      setCurrentBg((prev) => (prev + 1) % heroImages.length);
    }, 5000);
    return () => clearInterval(interval);
  }, []);

  return (
    <main className="premium-home">

      {/* Floating Transparent Logo synced with background */}
      <Link to="/" className="floating-logo">
        <svg width="32" height="32" viewBox="0 0 24 24" fill="none" xmlns="http://www.w3.org/2000/svg">
          <path d="M12 2L2 7L12 12L22 7L12 2Z" stroke="white" strokeWidth="2" strokeLinecap="round" strokeLinejoin="round"/>
          <path d="M2 17L12 22L22 17" stroke="white" strokeWidth="2" strokeLinecap="round" strokeLinejoin="round"/>
          <path d="M2 12L12 17L22 12" stroke="white" strokeWidth="2" strokeLinecap="round" strokeLinejoin="round"/>
        </svg>
        <span className="brand-text">Smart<span className="brand-highlight">SPACE</span></span>
```

app > src > Home.css > ...

```css
/* --- Global & Premium Theming --- */
:root {
  --bg-base: #050505;
  --text-main: #f3f3f3;
  --text-muted: #888888;
  --glass-bg: rgba(255, 255, 255, 0.03);
  --glass-border: rgba(255, 255, 255, 0.08);
  --ease-out-expo: cubic-bezier(0.16, 1, 0.3, 1);
}

body, html {
  background-color: var(--bg-base);
  color: var(--text-main);
  font-family: 'Inter', -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial, sans-serif;
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;
}

.premium-home {
  background-color: var(--bg-base);
  overflow-x: hidden;
  position: relative;
}

/* --- Floating Transparent Logo --- */
.floating-logo {
  position: absolute;
  top: 2rem;
  left: 2.5rem;
  z-index: 1000;
  display: flex;
  align-items: center;
  gap: 0.75rem;
  text-decoration: none;
  color: #ffffff;
  opacity: 0.5;
  transition: all 0.4s var(--ease-out-expo);
}

.floating-logo:hover {
  opacity: 1;
}

.floating-logo svg {
  transition: transform 0.4s ease;
}

.floating-logo:hover svg {
  transform: rotate(90deg);
```

app > src > Generate.jsx > ...

```jsx
import React, { useState } from 'react';
import axios from 'axios';
import { Container, Row, Col, Form, Button, Image, Spinner, Alert } from 'react-bootstrap';
import { Link } from 'react-router-dom';
import './Generate.css';
import defaultPreview from './assets/preview.jpg';

function Generate() {
  const [selectedFile, setSelectedFile] = useState(null);
  const [style, setStyle] = useState('');
  const [outputImage, setOutputImage] = useState(null);
  const [loading, setLoading] = useState(false);
  const [error, setError] = useState(null);
  const [isImageGenerated, setIsImageGenerated] = useState(false);

  const styles = ["modern", "bohemian", "industrial", "minimalist"];

  const handleFileChange = (event) => {
    setSelectedFile(event.target.files[0]);
    setOutputImage(null);
    setIsImageGenerated(false);
  };

  const handleStyleChange = (event) => {
    setStyle(event.target.value);
  };

  const handleSubmit = async (event) => {
    event.preventDefault();
    setOutputImage(null);
    setError(null);
    setIsImageGenerated(false);

    if (!selectedFile || !style) {
      setError("Please select both an image and a style.");
      return;
    }

    setLoading(true);

    const formData = new FormData();
    formData.append('file', selectedFile);
    formData.append('style', style);

    try {
      const response = await axios.post('http://127.0.0.1:5001/generate', formData, {
        headers: {
          'Content-Type': 'multipart/form-data',
```

app > src > Generate.css > ...

```css
/* --- Global Variables --- */
:root {
  --bg-base: #050505;
  --text-main: #f3f3f3;
  --text-muted: #888888;
  --glass-bg: rgba(255, 255, 255, 0.03);
  --glass-border: rgba(255, 255, 255, 0.08);
  --ease-out-expo: cubic-bezier(0.16, 1, 0.3, 1);
}

/* Base Layout */
.generate-background {
  background-color: var(--bg-base);
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  position: relative;
  font-family: 'Inter', -apple-system, BlinkMacSystemFont, sans-serif;
  color: var(--text-main);
  padding: 6rem 0;
  overflow-x: hidden;
}

/* Glowing AI Orb Background */
.generate-background::before {
  content: '';
  position: absolute;
  top: 50%;
  left: 50%;
  width: 60vw;
  height: 60vw;
  background: radial-gradient(circle, rgba(255,255,255,0.03) 0%, transparent 60%);
  transform: translate(-50%, -50%);
  z-index: 0;
  pointer-events: none;
}

.generate-container {
  position: relative;
  z-index: 2;
  width: 100%;
}

/* --- Floating Logo --- */
.floating-logo {
  position: absolute;
  top: 2rem;
  left: 2.5rem;
```

# CHAPTER 11

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